

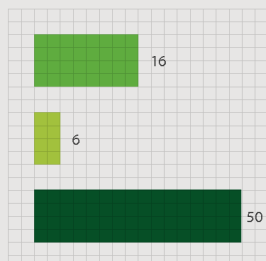
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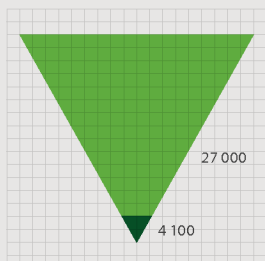


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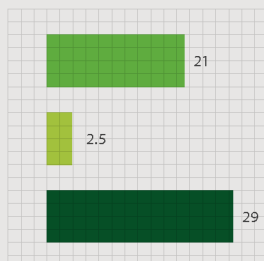
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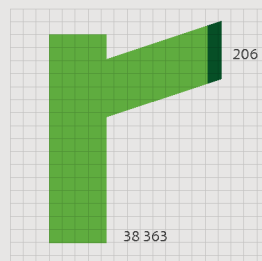
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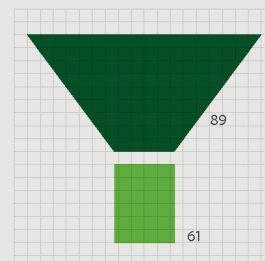
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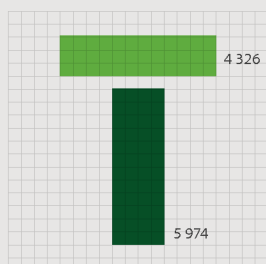
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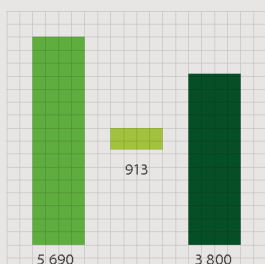
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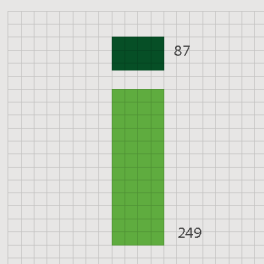
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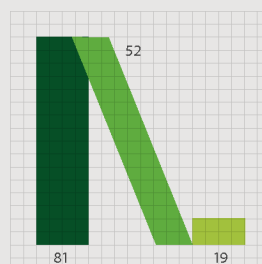
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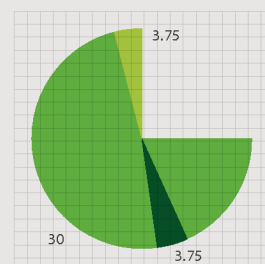
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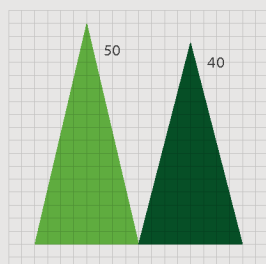
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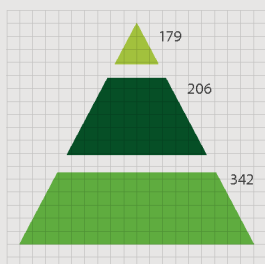
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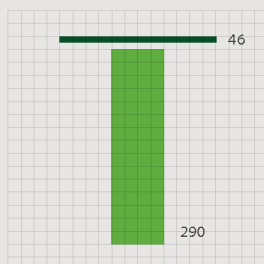
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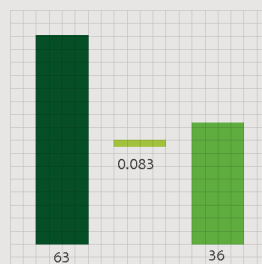
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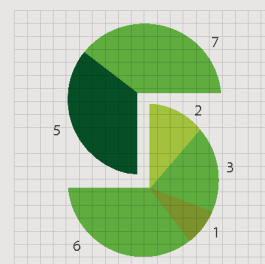
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EVERYTHING MATHS

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www.siyavula.com

info@siyavula.com

021 469 4771

Siyavula Authors

Alison Jenkin; Marina van Zyl; Dr. Carl Scheffler

Siyavula and DBE team

Neels van der Westhuizen; Leonard Gumani Mudau; Ewald Zietsman; Bridget Nash; Pertunia Mpho Letwaba; Josephine Mamaroke Phatlane; William Buthane Chauke; Nicola du Toit; Heather Williams

Siyavula and Free High School Science Text contributors

Dr. Mark Horner; Dr. Samuel Halliday; Dr. Sarah Blyth; Dr. Rory Adams; Dr. Spencer Wheaton

Iesrafeel Abbas; Sarah Abel; Dr. Rory Adams; Andrea Africa; Wiehan Agenbag; Matthew Amundsen; Ben Anhalt; Prashant Arora; Amos Baloyi; Bongani Baloyi; Raymond Barbour; Caro-Joy Barendse; Richard Baxter; Tara Beckerling; Tim van Beek; Mariaan Bester; Jennifer de Beyer; Dr. Sarah Blyth; Sebastian Bodenstein; Martin Bongers; Thinus Booysen; Gareth Boxall; Stephan Brandt; Hannes Breytenbach; Alexander Briell; Wilbur Britz; Graeme Broster; Craig Brown; Michail Brynard; Deanne de Bude; Richard Burge; Bianca B hmer; Jan Buys; George Calder-Potts; Eleanor Cameron; Mark Carolissen; Shane Carolissen; Richard Case; Sithembile Cele; Alice Chang; Richard Cheng; Fanny Cherblanc; Dr. Christine Chung; Brett Cocks; Roch  Compaan; Willem Conradie; Stefaan Conradie; Rocco Coppejans; Tim Craib; Andrew Craig; Tim Crombie; Dan Crytser; Jock Currie; Dr. Anne Dabrowski; Laura Daniels; Gareth Davies; Sandra Dickson; Sean Dobbs; Buhle Donga; William Donkin; Esmi Dreyer; Matthew Duddy; Christel Durie; Fernando Durrell; Dr. Dan Dwyer; Frans van Eeden; Alexander Ellis; Tom Ellis; Andrew Fisher; Giovanni Franzoni; Olivia Gillett; Ingrid von Glehn; Tamara von Glehn; Lindsay Glesener; Kevin Godby; Dr. Vanessa Godfrey; Terence Goldberg; Dr. Johan Gonzalez; Saaligha Gool; Hemant Gopal; Dr. Stephanie Gould; Umeshree Govender; Heather Gray; Lynn Greeff; Jaco Greyling; Martli Greyvenstein; Carine Grobbelaar; Suzanne Grov ; Dr. Tom Gutierrez; Brooke Haag; Kate Hadley; Alex Hall; Dr. Sam Halliday; Asheena Hanuman; Dr. Melanie Dymond Harper; Ebrahim Harris; Dr. Nicholas Harrison; Neil Hart; Nicholas Hatcher; Jason Hayden; Laura Hayward; Dr. William P. Heal; Pierre van Heerden; Dr. Fritha Hennessy; Dr. Colleen Henning; Shaun Hewitson; Millie Hilgart; Grant Hillebrand; Nick Hobbs; Chris Holdsworth; Dr. Benne Holwerda; Dr. Mark Horner; Robert Hovden; Mfandaidza Hove; Jennifer Hsieh; George Hugo; Laura Huss; Prof. Ed Jacobs

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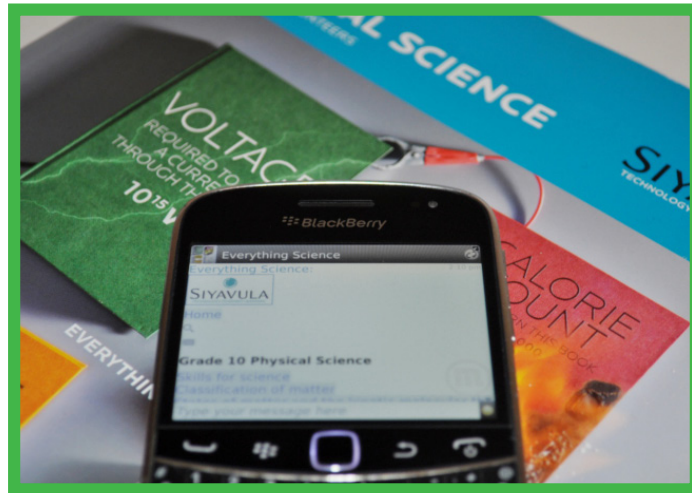
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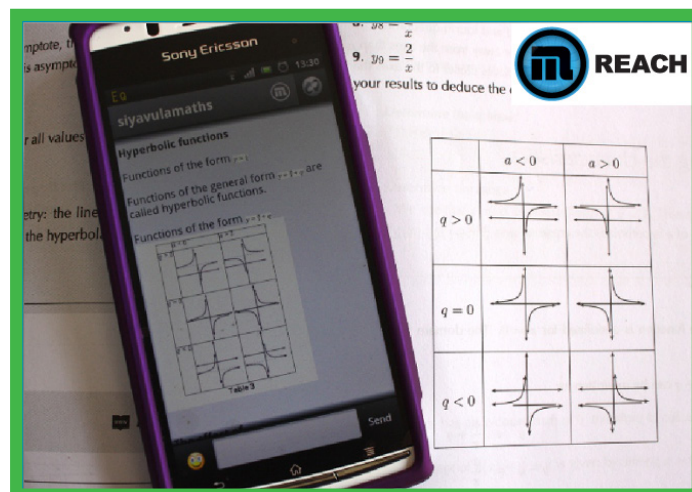
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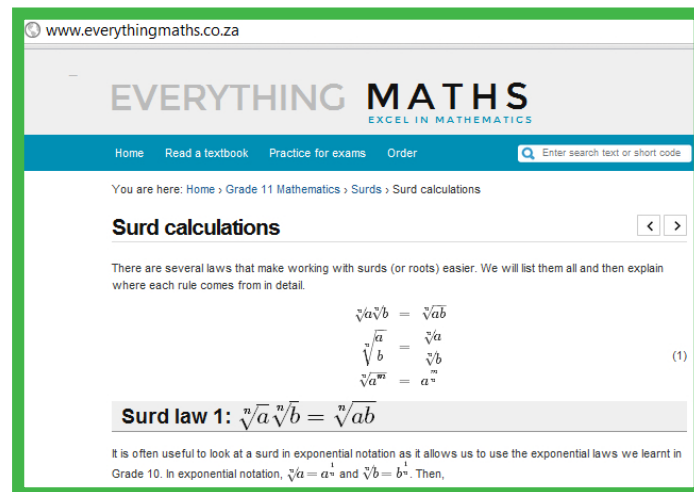


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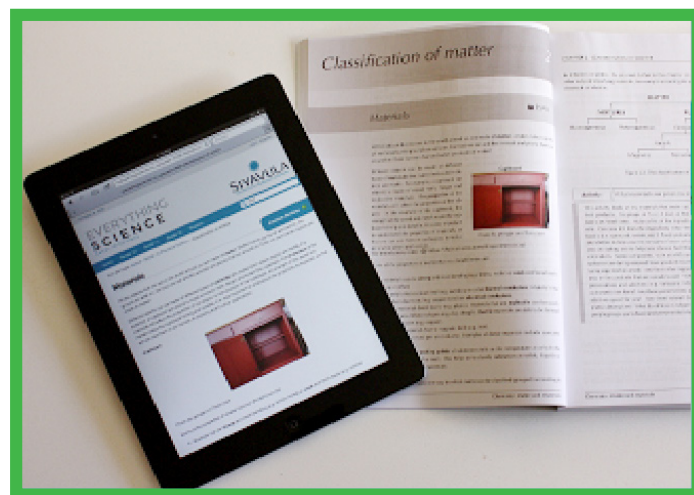
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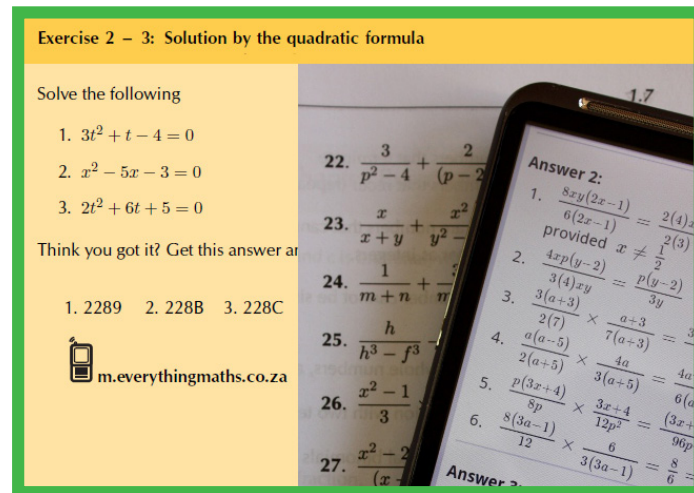


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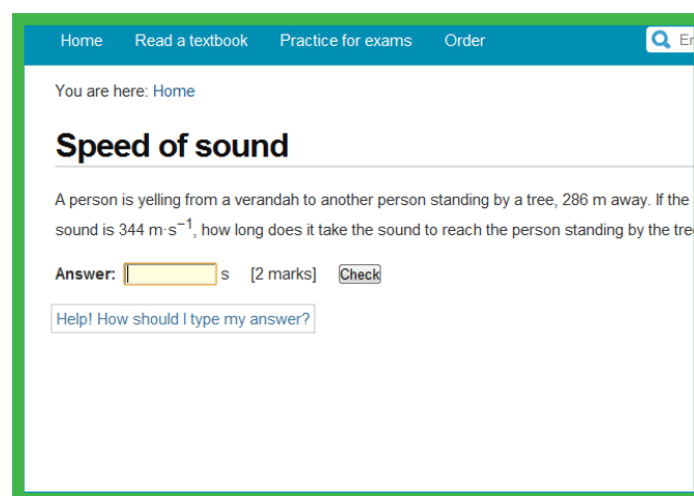
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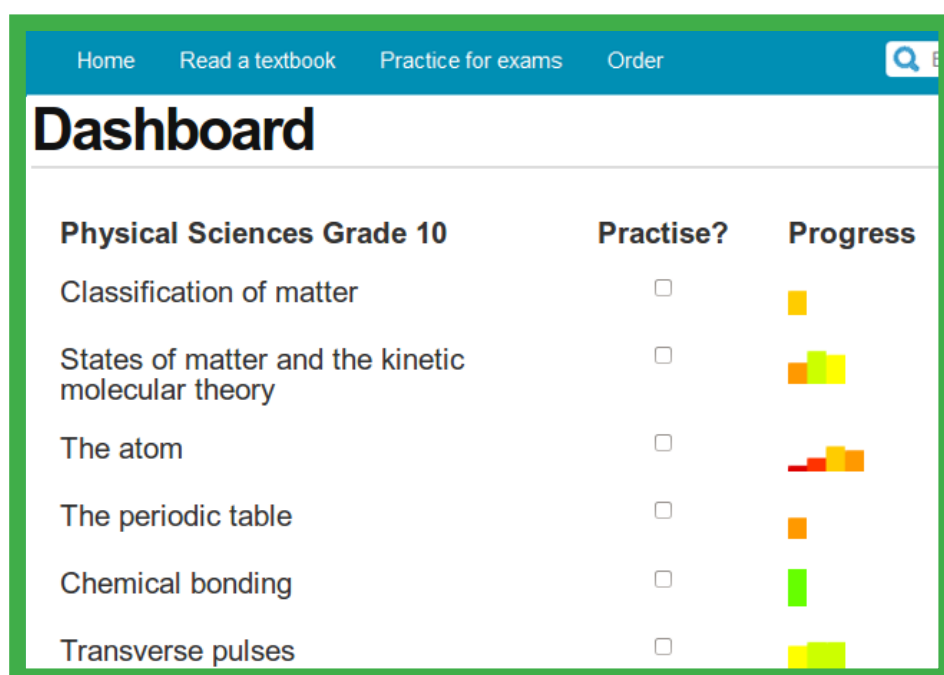


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EVERYTHING MATHS

Mathematics is commonly thought of as being about numbers but mathematics is actually a language! Mathematics is the language that nature speaks to us in. As we learn to understand and speak this language, we can discover many of nature's secrets. Just as understanding someone's language is necessary to learn more about them, mathematics is required to learn about all aspects of the world – whether it is physical sciences, life sciences or even finance and economics.

The great writers and poets of the world have the ability to draw on words and put them together in ways that can tell beautiful or inspiring stories. In a similar way, one can draw on mathematics to explain and create new things. Many of the modern technologies that have enriched our lives are greatly dependent on mathematics. DVDs, Google searches, bank cards with PIN numbers are just some examples. And just as words were not created specifically to tell a story but their existence enabled stories to be told, so the mathematics used to create these technologies was not developed for its own sake, but was available to be drawn on when the time for its application was right.

There is in fact not an area of life that is not affected by mathematics. Many of the most sought after careers depend on the use of mathematics. Civil engineers use mathematics to determine how to best design new structures; economists use mathematics to describe and predict how the economy will react to certain changes; investors use mathematics to price certain types of shares or calculate how risky particular investments are; software developers use mathematics for many of the algorithms (such as Google searches and data security) that make programmes useful.

But, even in our daily lives mathematics is everywhere – in our use of distance, time and money. Mathematics is even present in art, design and music as it informs proportions and musical tones. The greater our ability to understand mathematics, the greater our ability to appreciate beauty and everything in nature. Far from being just a cold and abstract discipline, mathematics embodies logic, symmetry, harmony and technological progress. More than any other language, mathematics is everywhere and universal in its application.

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Exponents and surds

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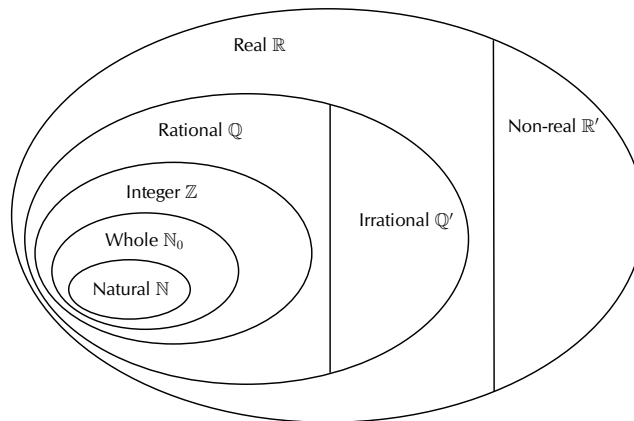
1.1 Revision

EMBF2

The number system

EMBF3

The diagram below shows the structure of the number system:



▶ See video: 2222 at www.everythingmaths.co.za

We use the following definitions:

- $\mathbb{N}$: natural numbers are $\{1; 2; 3; \dots\}$
- $\mathbb{N}_0$: whole numbers are $\{0; 1; 2; 3; \dots\}$
- $\mathbb{Z}$: integers are $\{\dots; -3; -2; -1; 0; 1; 2; 3; \dots\}$
- $\mathbb{Q}$: rational numbers are numbers which can be written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$, or as a terminating or recurring decimal number.
Examples: $-\frac{7}{2}$; $-2,25$; 0 ; $\sqrt{9}$; $0,8$; $\frac{23}{1}$
- $\mathbb{Q}'$: irrational numbers are numbers that cannot be written as a fraction with the numerator and denominator as integers. Irrational numbers also include decimal numbers that neither terminate nor recur.
Examples: $\sqrt{3}$; $\sqrt[5]{2}$; π ; $\frac{1+\sqrt{5}}{2}$; $1,27548\dots$
- $\mathbb{R}$: real numbers include all rational and irrational numbers.
- $\mathbb{R}'$: non-real numbers or imaginary numbers are numbers that are not real.
Examples: $\sqrt{-25}$; $\sqrt[4]{-1}$; $-\sqrt{-\frac{1}{16}}$

▶ See video: 2223 at www.everythingmaths.co.za

Exercise 1 – 1: The number system

Use the list of words below to describe each of the following numbers (in some cases multiple words will be applicable):

- Natural ($\mathbb{N}$)
- Whole ($\mathbb{N}_0$)
- Integer ($\mathbb{Z}$)
- Rational ($\mathbb{Q}$)
- Irrational ($\mathbb{Q}'$)
- Real ($\mathbb{R}$)
- Non-real ($\mathbb{R}'$)

1. $\sqrt{7}$
2. 0,01
3. $16\frac{2}{5}$
4. $\sqrt{6\frac{1}{4}}$
5. 0
6. 2π
7. $-5,3\dot{8}$
8. $\frac{1-\sqrt{2}}{2}$
9. $-\sqrt{-3}$
10. $(\pi)^2$
11. $-\frac{9}{11}$
12. $\sqrt[3]{-8}$
13. $\frac{22}{7}$
14. 2,45897...
15. $0,\overline{65}$
16. $\sqrt[5]{-32}$

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1. 2224 2. 2225 3. 2226 4. 2227 5. 2228 6. 2229
7. 222B 8. 222C 9. 222D 10. 222F 11. 222G 12. 222H
13. 222J 14. 222K 15. 222M 16. 222N



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Laws of exponents

EMBF4

We use exponential notation to show that a number or variable is multiplied by itself a certain number of times. The exponent, also called the index or power, indicates the number of times the multiplication is repeated.

base $\leftarrow a^n \rightarrow$ exponent/index

$$a^n = a \times a \times a \times \dots \times a \quad (n \text{ times}) \quad (a \in \mathbb{R}, n \in \mathbb{N})$$

▶ See video: 222P at www.everythingmaths.co.za

Examples:

1. $2 \times 2 \times 2 \times 2 = 2^4$
2. $0,71 \times 0,71 \times 0,71 = (0,71)^3$
3. $(501)^2 = 501 \times 501$
4. $k^6 = k \times k \times k \times k \times k \times k$

For x^2 , we say x is squared and for y^3 , we say that y is cubed. In the last example we have k^6 ; we say that k is raised to the sixth power.

We also have the following definitions for exponents. It is important to remember that we always write the final answer with a positive exponent.

- $a^0 = 1$ ($a \neq 0$ because 0^0 is undefined)
- $a^{-n} = \frac{1}{a^n}$ ($a \neq 0$ because $\frac{1}{0}$ is undefined)

Examples:

1. $5^{-2} = \frac{1}{5^2} = \frac{1}{25}$
2. $(-36)^0 x = (1)x = x$
3. $\frac{7p^{-1}}{q^3 t^{-2}} = \frac{7t^2}{pq^3}$

We use the following laws for working with exponents:

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- $(a^m)^n = a^{mn}$

where $a > 0$, $b > 0$ and $m, n \in \mathbb{Z}$.

Worked example 1: Laws of exponents

QUESTION

Simplify the following:

1. $5(m^{2t})^p \times 2(m^{3p})^t$
2. $\frac{8k^3 x^2}{(xk)^2}$

$$3. \frac{2^2 \times 3 \times 7^4}{(7 \times 2)^4}$$

$$4. 3(3^b)^a$$

SOLUTION

$$1. 5(m^{2t})^p \times 2(m^{3p})^t = 10m^{2pt+3pt} = 10m^{5pt}$$

$$2. \frac{8k^3x^2}{(xk)^2} = \frac{8k^3x^2}{x^2k^2} = 8k^{(3-2)}x^{(2-2)} = 8k^1x^0 = 8k$$

$$3. \frac{2^2 \times 3 \times 7^4}{(7 \times 2)^4} = \frac{2^2 \times 3 \times 7^4}{7^4 \times 2^4} = 2^{(2-4)} \times 3 \times 7^{(4-4)} = 2^{-2} \times 3 = \frac{3}{4}$$

$$4. 3(3^b)^a = 3 \times 3^{ab} = 3^{ab+1}$$

Worked example 2: Laws of exponents

QUESTION

Simplify: $\frac{3^m - 3^{m+1}}{4 \times 3^m - 3^m}$

SOLUTION

Step 1: Simplify to a form that can be factorised

$$\frac{3^m - 3^{m+1}}{4 \times 3^m - 3^m} = \frac{3^m - (3^m \times 3)}{4 \times 3^m - 3^m}$$

Step 2: Take out a common factor

$$= \frac{3^m(1 - 3)}{3^m(4 - 1)}$$

Step 3: Cancel the common factor and simplify

$$\begin{aligned} &= \frac{1 - 3}{4 - 1} \\ &= -\frac{2}{3} \end{aligned}$$

Exercise 1 – 2: Laws of exponents

Simplify the following:

1. $4 \times 4^{2a} \times 4^2 \times 4^a$

2. $\frac{3^2}{2^{-3}}$

3. $(3p^5)^2$

4. $\frac{k^2 k^{3x-4}}{k^x}$

5. $(5^{z-1})^2 + 5^z$

6. $(\frac{1}{4})^0$

7. $(x^2)^5$

8. $(\frac{a}{b})^{-2}$

9. $(m+n)^{-1}$

10. $2(p^t)^s$

11. $\frac{1}{(\frac{1}{a})^{-1}}$

12. $\frac{k^0}{k^{-1}}$

13. $\frac{-2}{-2^{-a}}$

14. $\frac{-h}{(-h)^{-3}}$

15. $\left(\frac{a^2 b^3}{c^3 d}\right)^2$

16. $10^7(7^0) \times 10^{-6}(-6)^0 - 6$

17. $m^3 n^2 \div nm^2 \times \frac{mn}{2}$

18. $(2^{-2} - 5^{-1})^{-2}$

19. $(y^2)^{-3} \div \left(\frac{x^2}{y^3}\right)^{-1}$

20. $\frac{2^{c-5}}{2^{c-8}}$

21. $\frac{2^{9a} \times 4^{6a} \times 2^2}{8^{5a}}$

22. $\frac{20t^5 p^{10}}{10t^4 p^9}$

23. $\left(\frac{9q^{-2s}}{q^{-3s}y^{-4a-1}}\right)^2$

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- | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1. 222R | 2. 222S | 3. 222T | 4. 222V | 5. 222W | 6. 222X |
| 7. 222Y | 8. 222Z | 9. 2232 | 10. 2233 | 11. 2234 | 12. 2235 |
| 13. 2236 | 14. 2237 | 15. 2238 | 16. 2239 | 17. 223B | 18. 223C |
| 19. 223D | 20. 223F | 21. 223G | 22. 223H | 23. 223J | |



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► See video: [222Q](#) at www.everythingmaths.co.za

1.2 Rational exponents and surds

EMBF5

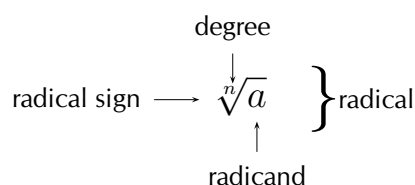
The laws of exponents can also be extended to include the rational numbers. A rational number is any number that can be written as a fraction with an integer in the numerator and in the denominator. We also have the following definitions for working with rational exponents.

- If $r^n = a$, then $r = \sqrt[n]{a}$ ($n \geq 2$)
- $a^{\frac{1}{n}} = \sqrt[n]{a}$
- $a^{-\frac{1}{n}} = (a^{-1})^{\frac{1}{n}} = \sqrt[n]{\frac{1}{a}}$
- $a^{\frac{m}{n}} = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m}$

where $a > 0$, $r > 0$ and $m, n \in \mathbb{Z}$, $n \neq 0$.

For $\sqrt{25} = 5$, we say that 5 is the square root of 25 and for $\sqrt[3]{8} = 2$, we say that 2 is the cube root of 8. For $\sqrt[5]{32} = 2$, we say that 2 is the fifth root of 32.

When dealing with exponents, a root refers to a number that is repeatedly multiplied by itself a certain number of times to get another number. A radical refers to a number written as shown below.



► See video: 223K at www.everythingmaths.co.za

The radical symbol and degree show which root is being determined. The radicand is the number under the radical symbol.

- If n is an even natural number, then the radicand must be positive, otherwise the roots are not real. For example, $\sqrt[4]{16} = 2$ since $2 \times 2 \times 2 \times 2 = 16$, but the roots of $\sqrt[4]{-16}$ are not real since $(-2) \times (-2) \times (-2) \times (-2) \neq -16$.
- If n is an odd natural number, then the radicand can be positive or negative. For example, $\sqrt[3]{27} = 3$ since $3 \times 3 \times 3 = 27$ and we can also determine $\sqrt[3]{-27} = -3$ since $(-3) \times (-3) \times (-3) = -27$.

It is also possible for there to be more than one n^{th} root of a number. For example, $(-2)^2 = 4$ and $2^2 = 4$, so both -2 and 2 are square roots of 4.

A surd is a radical which results in an irrational number. Irrational numbers are numbers that cannot be written as a fraction with the numerator and the denominator as integers. For example, $\sqrt{12}$, $\sqrt[3]{100}$, $\sqrt[5]{25}$ are surds.

Worked example 3: Rational exponents

QUESTION

Write each of the following as a radical and simplify where possible:

1. $18^{\frac{1}{2}}$
2. $(-125)^{-\frac{1}{3}}$

3. $4^{\frac{3}{2}}$

4. $(-81)^{\frac{1}{2}}$

5. $(0,008)^{\frac{1}{3}}$

SOLUTION

1. $18^{\frac{1}{2}} = \sqrt{18}$

2. $(-125)^{-\frac{1}{3}} = \sqrt[3]{(-125)^{-1}} = \sqrt[3]{\frac{1}{-125}} = \sqrt[3]{\frac{1}{(-5)^3}} = -\frac{1}{5}$

3. $4^{\frac{3}{2}} = (4^3)^{\frac{1}{2}} = \sqrt{4^3} = \sqrt{64} = 8$

4. $(-81)^{\frac{1}{2}} = \sqrt{-81} = \text{not real}$

5. $(0,008)^{\frac{1}{3}} = \sqrt[3]{\frac{8}{1000}} = \sqrt[3]{\frac{2^3}{10^3}} = \frac{2}{10} = \frac{1}{5}$

📺 See video: [223M](https://www.everythingmaths.co.za) at www.everythingmaths.co.za

Worked example 4: Rational exponents

QUESTION

Simplify without using a calculator:

$$\left(\frac{5}{4^{-1} - 9^{-1}} \right)^{\frac{1}{2}}$$

SOLUTION

Step 1: Write the fraction with positive exponents in the denominator

$$\left(\frac{5}{\frac{1}{4} - \frac{1}{9}} \right)^{\frac{1}{2}}$$

Step 2: Simplify the denominator

$$\begin{aligned}
&= \left(\frac{5}{\frac{9-4}{36}} \right)^{\frac{1}{2}} \\
&= \left(\frac{5}{\frac{5}{36}} \right)^{\frac{1}{2}} \\
&= \left(5 \div \frac{5}{36} \right)^{\frac{1}{2}} \\
&= \left(5 \times \frac{36}{5} \right)^{\frac{1}{2}} \\
&= (36)^{\frac{1}{2}}
\end{aligned}$$

Step 3: Take the square root

$$\begin{aligned}
&= \sqrt{36} \\
&= 6
\end{aligned}$$

Exercise 1 – 3: Rational exponents and surds

1. Simplify the following and write answers with positive exponents:

a) $\sqrt{49}$

b) $\sqrt{36^{-1}}$

c) $\sqrt[3]{6^{-2}}$

d) $\sqrt[3]{-\frac{64}{27}}$

e) $\sqrt[4]{(16x^4)^3}$

2. Simplify:

a) $s^{\frac{1}{2}} \div s^{\frac{1}{3}}$

b) $(64m^6)^{\frac{2}{3}}$

c) $\frac{12m^{\frac{7}{9}}}{8m^{-\frac{11}{9}}}$

d) $(5x)^0 + 5x^0 - (0,25)^{-0,5} + 8^{\frac{2}{3}}$

3. Use the laws to re-write the following expression as a power of x :

$$x\sqrt{x\sqrt{x\sqrt{x\sqrt{x}}}}$$

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1a. 223P 1b. 223Q 1c. 223R 1d. 223S 1e. 223T 2a. 223V
 2b. 223W 2c. 223X 2d. 223Y 3. 223Z



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We have seen in previous examples and exercises that rational exponents are closely related to surds. It is often useful to write a surd in exponential notation as it allows us to use the exponential laws.

The additional laws listed below make simplifying surds easier:

- $\sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$
- $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$
- $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$
- $\sqrt[n]{a^m} = a^{\frac{m}{n}}$
- $(\sqrt[n]{a})^m = a^{\frac{m}{n}}$

► See video: 223N at www.everythingmaths.co.za

Worked example 5: Simplifying surds

QUESTION

Show that:

1. $\sqrt[n]{a} \times \sqrt[n]{b} = \sqrt[n]{ab}$
2. $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$

SOLUTION

1.

$$\begin{aligned}\sqrt[n]{a} \times \sqrt[n]{b} &= a^{\frac{1}{n}} \times b^{\frac{1}{n}} \\ &= (ab)^{\frac{1}{n}} \\ &= \sqrt[n]{ab}\end{aligned}$$

2.

$$\begin{aligned}\sqrt[n]{\frac{a}{b}} &= \left(\frac{a}{b}\right)^{\frac{1}{n}} \\ &= \frac{a^{\frac{1}{n}}}{b^{\frac{1}{n}}} \\ &= \frac{\sqrt[n]{a}}{\sqrt[n]{b}}\end{aligned}$$

Examples:

$$1. \sqrt{2} \times \sqrt{32} = \sqrt{2 \times 32} = \sqrt{64} = 8$$

$$2. \frac{\sqrt[3]{24}}{\sqrt[3]{3}} = \sqrt[3]{\frac{24}{3}} = \sqrt[3]{8} = 2$$

$$3. \sqrt{\sqrt{81}} = \sqrt[4]{81} = \sqrt[4]{3^4} = 3$$

Like and unlike surds

EMBF7

Two surds $\sqrt[n]{a}$ and $\sqrt[n]{b}$ are like surds if $m = n$, otherwise they are called unlike surds. For example, $\sqrt{\frac{1}{3}}$ and $-\sqrt{61}$ are like surds because $m = n = 2$. Examples of unlike surds are $\sqrt[3]{5}$ and $\sqrt[5]{7y^3}$ since $m \neq n$.

Simplest surd form

EMBF8

We can sometimes simplify surds by writing the radicand as a product of factors that can be further simplified using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$.

► See video: 2242 at www.everythingmaths.co.za

Worked example 6: Simplest surd form

QUESTION

Write the following in simplest surd form: $\sqrt{50}$

SOLUTION

Step 1: Write the radicand as a product of prime factors

$$\begin{aligned}\sqrt{50} &= \sqrt{5 \times 5 \times 2} \\ &= \sqrt{5^2 \times 2}\end{aligned}$$

Step 2: Simplify using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

$$\begin{aligned}&= \sqrt{5^2} \times \sqrt{2} \\ &= 5 \times \sqrt{2} \\ &= 5\sqrt{2}\end{aligned}$$

Sometimes a surd cannot be simplified. For example, $\sqrt{6}$, $\sqrt[3]{30}$ and $\sqrt[4]{42}$ are already in their simplest form.

Worked example 7: Simplest surd form

QUESTION

Write the following in simplest surd form: $\sqrt[3]{54}$

SOLUTION

Step 1: Write the radicand as a product of prime factors

$$\begin{aligned}\sqrt[3]{54} &= \sqrt[3]{3 \times 3 \times 3 \times 2} \\ &= \sqrt[3]{3^3 \times 2}\end{aligned}$$

Step 2: Simplify using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

$$\begin{aligned}&= \sqrt[3]{3^3} \times \sqrt[3]{2} \\ &= 3 \times \sqrt[3]{2} \\ &= 3\sqrt[3]{2}\end{aligned}$$

Worked example 8: Simplest surd form

QUESTION

Simplify: $\sqrt{147} + \sqrt{108}$

SOLUTION

Step 1: Write the radicands as a product of prime factors

$$\begin{aligned}\sqrt{147} + \sqrt{108} &= \sqrt{49 \times 3} + \sqrt{36 \times 3} \\ &= \sqrt{7^2 \times 3} + \sqrt{6^2 \times 3}\end{aligned}$$

Step 2: Simplify using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

$$\begin{aligned}
&= (\sqrt{7^2} \times \sqrt{3}) + (\sqrt{6^2} \times \sqrt{3}) \\
&= (7 \times \sqrt{3}) + (6 \times \sqrt{3}) \\
&= 7\sqrt{3} + 6\sqrt{3}
\end{aligned}$$

Step 3: Simplify and write the final answer

$$13\sqrt{3}$$

Worked example 9: Simplest surd form

QUESTION

Simplify: $(\sqrt{20} - \sqrt{5})^2$

SOLUTION

Step 1: Factorise the radicands where possible

$$(\sqrt{20} - \sqrt{5})^2 = (\sqrt{4 \times 5} - \sqrt{5})^2$$

Step 2: Simplify using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

$$\begin{aligned}
&= (\sqrt{4} \times \sqrt{5} - \sqrt{5})^2 \\
&= (2 \times \sqrt{5} - \sqrt{5})^2 \\
&= (2\sqrt{5} - \sqrt{5})^2
\end{aligned}$$

Step 3: Simplify and write the final answer

$$\begin{aligned}
&= (\sqrt{5})^2 \\
&= 5
\end{aligned}$$

Worked example 10: Simplest surd form with fractions

QUESTION

Write in simplest surd form: $\sqrt{75} \times \sqrt[3]{(48)^{-1}}$

SOLUTION

Step 1: Factorise the radicands where possible

$$\begin{aligned}\sqrt{75} \times \sqrt[3]{(48)^{-1}} &= \sqrt{25 \times 3} \times \sqrt[3]{\frac{1}{48}} \\ &= \sqrt{25 \times 3} \times \frac{1}{\sqrt[3]{8 \times 6}}\end{aligned}$$

Step 2: Simplify using $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

$$\begin{aligned}&= \sqrt{25} \times \sqrt{3} \times \frac{1}{\sqrt[3]{8} \times \sqrt[3]{6}} \\ &= 5 \times \sqrt{3} \times \frac{1}{2 \times \sqrt[3]{6}}\end{aligned}$$

Step 3: Simplify and write the final answer

$$\begin{aligned}&= 5\sqrt{3} \times \frac{1}{2\sqrt[3]{6}} \\ &= \frac{5\sqrt{3}}{2\sqrt[3]{6}}\end{aligned}$$

Exercise 1 – 4: Simplification of surds

1. Simplify the following and write answers with positive exponents:

a) $\sqrt[3]{16} \times \sqrt[3]{4}$

b) $\sqrt{a^2b^3} \times \sqrt{b^5c^4}$

c) $\frac{\sqrt{12}}{\sqrt{3}}$

d) $\sqrt{x^2y^{13}} \div \sqrt{y^5}$

2. Simplify the following:

a) $\left(\frac{1}{a} - \frac{1}{b}\right)^{-1}$

b) $\frac{b-a}{a^{\frac{1}{2}} - b^{\frac{1}{2}}}$

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1a. 2243 1b. 2244 1c. 2245 1d. 2246 2a. 2247 2b. 2248



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Rationalising denominators

EMBF9

It is often easier to work with fractions that have rational denominators instead of surd denominators. By rationalising the denominator, we convert a fraction with a surd in the denominator to a fraction that has a rational denominator.

Worked example 11: Rationalising the denominator

QUESTION

Rationalise the denominator:

$$\frac{5x-16}{\sqrt{x}}$$

SOLUTION

Step 1: Multiply the fraction by $\frac{\sqrt{x}}{\sqrt{x}}$

Notice that $\frac{\sqrt{x}}{\sqrt{x}} = 1$, so the value of the fraction has not been changed.

$$\frac{5x-16}{\sqrt{x}} \times \frac{\sqrt{x}}{\sqrt{x}} = \frac{\sqrt{x}(5x-16)}{\sqrt{x} \times \sqrt{x}}$$

Step 2: Simplify the denominator

$$\begin{aligned} &= \frac{\sqrt{x}(5x-16)}{(\sqrt{x})^2} \\ &= \frac{\sqrt{x}(5x-16)}{x} \end{aligned}$$

The term in the denominator has changed from a surd to a rational number. Expressing the surd in the numerator is the preferred way of writing expressions.

Worked example 12: Rationalising the denominator

QUESTION

Write the following with a rational denominator:

$$\frac{y - 25}{\sqrt{y} + 5}$$

SOLUTION

Step 1: Multiply the fraction by $\frac{\sqrt{y}-5}{\sqrt{y}-5}$

To eliminate the surd from the denominator, we must multiply the fraction by an expression that will result in a difference of two squares in the denominator.

$$\frac{y - 25}{\sqrt{y} + 5} \times \frac{\sqrt{y} - 5}{\sqrt{y} - 5}$$

Step 2: Simplify the denominator

$$\begin{aligned} &= \frac{(y - 25)(\sqrt{y} - 5)}{(\sqrt{y} + 5)(\sqrt{y} - 5)} \\ &= \frac{(y - 25)(\sqrt{y} - 5)}{(\sqrt{y})^2 - 25} \\ &= \frac{(y - 25)(\sqrt{y} - 5)}{y - 25} \\ &= \sqrt{y} - 5 \end{aligned}$$

► See video: 2249 at www.everythingmaths.co.za

Exercise 1 – 5: Rationalising the denominator

Rationalise the denominator in each of the following:

1. $\frac{10}{\sqrt{5}}$

6. $\frac{\sqrt{3} + \sqrt{7}}{\sqrt{2}}$

2. $\frac{3}{\sqrt{6}}$

7. $\frac{3\sqrt{p} - 4}{\sqrt{p}}$

3. $\frac{2}{\sqrt{3}} \div \frac{\sqrt{2}}{3}$

8. $\frac{t - 4}{\sqrt{t} + 2}$

4. $\frac{3}{\sqrt{5} - 1}$

9. $(1 + \sqrt{m})^{-1}$

5. $\frac{x}{\sqrt{y}}$

10. $a(\sqrt{a} \div \sqrt{b})^{-1}$

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1. 224B 2. 224C 3. 224D 4. 224F 5. 224G 6. 224H
7. 224J 8. 224K 9. 224M 10. 224N



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1.3 Solving surd equations

EMBFB

We also need to be able to solve equations that involve surds.

▶ See video: 224P at www.everythingmaths.co.za

Worked example 13: Surd equations

QUESTION

Solve for x : $5\sqrt[3]{x^4} = 405$

SOLUTION

Step 1: Write in exponential notation

$$\begin{aligned}5(x^4)^{\frac{1}{3}} &= 405 \\5x^{\frac{4}{3}} &= 405\end{aligned}$$

Step 2: Divide both sides of the equation by 5 and simplify

$$\begin{aligned}\frac{5x^{\frac{4}{3}}}{5} &= \frac{405}{5} \\x^{\frac{4}{3}} &= 81 \\x^{\frac{4}{3}} &= 3^4\end{aligned}$$

Step 3: Simplify the exponents

$$\begin{aligned}\left(x^{\frac{4}{3}}\right)^{\frac{3}{4}} &= \left(3^4\right)^{\frac{3}{4}} \\x &= 3^3 \\x &= 27\end{aligned}$$

Step 4: Check the solution by substituting the answer back into the original equation

$$\begin{aligned}\text{LHS} &= 5\sqrt[3]{x^4} \\ &= 5(27)^{\frac{4}{3}} \\ &= 5(3^3)^{\frac{4}{3}} \\ &= 5(3^4) \\ &= 405 \\ &= \text{RHS}\end{aligned}$$

Worked example 14: Surd equations

QUESTION

Solve for z : $z - 4\sqrt{z} + 3 = 0$

SOLUTION

Step 1: Factorise

$$\begin{aligned}z - 4\sqrt{z} + 3 &= 0 \\ z - 4z^{\frac{1}{2}} + 3 &= 0 \\ (z^{\frac{1}{2}} - 3)(z^{\frac{1}{2}} - 1) &= 0\end{aligned}$$

Step 2: Solve for both factors

The zero law states: if $a \times b = 0$, then $a = 0$ or $b = 0$.

$$\therefore (z^{\frac{1}{2}} - 3) = 0 \text{ or } (z^{\frac{1}{2}} - 1) = 0$$

Therefore

$$\begin{aligned}z^{\frac{1}{2}} - 3 &= 0 \\ z^{\frac{1}{2}} &= 3 \\ \left(z^{\frac{1}{2}}\right)^2 &= 3^2 \\ z &= 9\end{aligned}$$

or

$$\begin{aligned}z^{\frac{1}{2}} - 1 &= 0 \\z^{\frac{1}{2}} &= 1 \\(z^{\frac{1}{2}})^2 &= 1^2 \\z &= 1\end{aligned}$$

Step 3: Check the solution by substituting both answers back into the original equation

If $z = 9$:

$$\begin{aligned}\text{LHS} &= z - 4\sqrt{z} + 3 \\&= 9 - 4\sqrt{9} + 3 \\&= 12 - 12 \\&= 0 \\&= \text{RHS}\end{aligned}$$

If $z = 1$:

$$\begin{aligned}\text{LHS} &= z - 4\sqrt{z} + 3 \\&= 1 - 4\sqrt{1} + 3 \\&= 4 - 4 \\&= 0 \\&= \text{RHS}\end{aligned}$$

Step 4: Write the final answer

The solution to $z - 4\sqrt{z} + 3 = 0$ is $z = 9$ or $z = 1$.

Worked example 15: Surd equations

QUESTION

Solve for p : $\sqrt{p-2} - 3 = 0$

SOLUTION

Step 1: Write the equation with only the square root on the left hand side

Use the additive inverse to get all other terms on the right hand side and only the

square root on the left hand side.

$$\sqrt{p-2} = 3$$

Step 2: Square both sides of the equation

$$\left(\sqrt{p-2}\right)^2 = 3^2$$

$$p-2 = 9$$

$$p = 11$$

Step 3: Check the solution by substituting the answer back into the original equation

If $p = 11$:

$$\text{LHS} = \sqrt{p-2} - 3$$

$$= \sqrt{11-2} - 3$$

$$= \sqrt{9} - 3$$

$$= 3 - 3$$

$$= 0$$

$$= \text{RHS}$$

Step 4: Write the final answer

The solution to $\sqrt{p-2} - 3 = 0$ is $p = 11$.

Exercise 1 – 6: Solving surd equations

Solve for the unknown variable (remember to check that the solution is valid):

1. $2^{x+1} - 32 = 0$

6. $x^{\frac{1}{3}}(x^{\frac{1}{3}} + 1) = 6$

2. $125(3^p) = 27(5^p)$

7. $2^{4n} - \frac{1}{\sqrt[4]{16}} = 0$

3. $2y^{\frac{1}{2}} - 3y^{\frac{1}{4}} + 1 = 0$

8. $\sqrt{31-10d} = 4-d$

4. $t-1 = \sqrt{7-t}$

9. $y - 10\sqrt{y} + 9 = 0$

5. $2z - 7\sqrt{z} + 3 = 0$

10. $f = 2 + \sqrt{19-2f}$

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1. 224Q 2. 224R 3. 224S 4. 224T 5. 224V 6. 224W
7. 224X 8. 224Y 9. 224Z 10. 2252



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There are many real world applications that require exponents. For example, exponentials are used to determine population growth and they are also used in finance to calculate different types of interest.

Worked example 16: Applications of exponentials**QUESTION**

A type of bacteria has a very high exponential growth rate at 80% every hour. If there are 10 bacteria, determine how many there will be in five hours, in one day and in one week?

SOLUTION**Step 1: Exponential formula**

$$\text{final population} = \text{initial population} \times (1 + \text{growth percentage})^{\text{time period in hours}}$$

Therefore, in this case:

$$\text{final population} = 10 (1,8)^n$$

where n = number of hours.

Step 2: In 5 hours

$$\text{final population} = 10 (1,8)^5 \approx 189$$

Step 3: In 1 day = 24 hours

$$\text{final population} = 10 (1,8)^{24} \approx 13\,382\,588$$

Step 4: In 1 week = 168 hours

$$\text{final population} = 10 (1,8)^{168} \approx 7,687 \times 10^{43}$$

Note this answer is given in scientific notation as it is a very big number.

Worked example 17: Applications of exponentials

QUESTION

A species of extremely rare deep water fish has a very long lifespan and rarely has offspring. If there are a total of 821 of this type of fish and their growth rate is 2% each month, how many will there be in half of a year? What will the population be in ten years and in one hundred years?

SOLUTION

Step 1: Exponential formula

$$\text{final population} = \text{initial population} \times (1 + \text{growth percentage})^{\text{time period in months}}$$

Therefore, in this case:

$$\text{final population} = 821(1,02)^n$$

where n = number of months.

Step 2: In half a year = 6 months

$$\text{final population} = 821(1,02)^6 \approx 925$$

Step 3: In 10 years = 120 months

$$\text{final population} = 821(1,02)^{120} \approx 8838$$

Step 4: In 100 years = 1200 months

$$\text{final population} = 821(1,02)^{1200} \approx 1,716 \times 10^{13}$$

Note this answer is also given in scientific notation as it is a very big number.

Exercise 1 – 7: Applications of exponentials

1. Ngobani invests R 5530 into an account which pays out a lump sum at the end of 6 years. If he gets R 9622,20 at the end of the period, what compound interest rate did the bank offer him? Give answer correct to one decimal place.
2. The current population of Johannesburg is 3 885 840 and the average rate of population growth in South Africa is 0,7% p.a. What can city planners expect the population of Johannesburg to be in 13 years time?
3. Abiona places 3 books in a stack on her desk. The next day she counts the books in the stack and then adds the same number of books to the top of the stack. After how many days will she have a stack of 192 books?

4. A type of mould has a very high exponential growth rate of 40% every hour. If there are initially 45 individual mould cells in the population, determine how many there will be in 19 hours.

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1.5 Summary

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▶ See presentation: 2257 at www.everythingmaths.co.za

1. The number system:

- $\mathbb{N}$: natural numbers are $\{1; 2; 3; \dots\}$
- $\mathbb{N}_0$: whole numbers are $\{0; 1; 2; 3; \dots\}$
- $\mathbb{Z}$: integers are $\{\dots; -3; -2; -1; 0; 1; 2; 3; \dots\}$
- $\mathbb{Q}$: rational numbers are numbers which can be written as $\frac{a}{b}$ where a and b are integers and $b \neq 0$, or as a terminating or recurring decimal number.
- $\mathbb{Q}'$: irrational numbers are numbers that cannot be written as a fraction with the numerator and denominator as integers. Irrational numbers also include decimal numbers that neither terminate nor recur.
- $\mathbb{R}$: real numbers include all rational and irrational numbers.
- $\mathbb{R}'$: non-real numbers or imaginary numbers are numbers that are not real.

2. Definitions:

- $a^n = a \times a \times a \times \dots \times a$ (n times) ($a \in \mathbb{R}, n \in \mathbb{N}$)
- $a^0 = 1$ ($a \neq 0$ because 0^0 is undefined)
- $a^{-n} = \frac{1}{a^n}$ ($a \neq 0$ because $\frac{1}{0}$ is undefined)

3. Laws of exponents:

- $a^m \times a^n = a^{m+n}$
- $\frac{a^m}{a^n} = a^{m-n}$
- $(ab)^n = a^n b^n$
- $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$
- $(a^m)^n = a^{mn}$

where $a > 0, b > 0$ and $m, n \in \mathbb{Z}$.

4. Rational exponents and surds:

- If $r^n = a$, then $r = \sqrt[n]{a}$ ($n \geq 2$)
- $a^{\frac{1}{n}} = \sqrt[n]{a}$
- $a^{-\frac{1}{n}} = (a^{-1})^{\frac{1}{n}} = \sqrt[n]{\frac{1}{a}}$
- $a^{\frac{m}{n}} = (a^m)^{\frac{1}{n}} = \sqrt[n]{a^m}$

where $a > 0$, $r > 0$ and $m, n \in \mathbb{Z}$, $n \neq 0$.

5. Simplification of surds:

- $\sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$
- $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$
- $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

Exercise 1 – 8: End of chapter exercises

1. Simplify as far as possible:

- a) $8^{-\frac{2}{3}}$
- b) $\sqrt{16} + 8^{-\frac{2}{3}}$

2. Simplify:

- a) $(x^3)^{\frac{4}{3}}$
- b) $(s^2)^{\frac{1}{2}}$
- c) $(m^5)^{\frac{5}{3}}$
- d) $(-m^2)^{\frac{4}{3}}$
- e) $-(m^2)^{\frac{4}{3}}$
- f) $(3y^{\frac{4}{3}})^4$

3. Simplify the following:

- a) $\frac{3a^{-2}b^{15}c^{-5}}{(a^{-4}b^3c)^{-\frac{5}{2}}}$
- b) $(9a^6b^4)^{\frac{1}{2}}$
- c) $(a^{\frac{3}{2}}b^{\frac{3}{4}})^{16}$
- d) $x^3\sqrt{x}$
- e) $\sqrt[3]{x^4b^5}$

4. Re-write the following expression as a power of x :

$$\frac{x\sqrt{x\sqrt{x\sqrt{x\sqrt{x}}}}}{x^2}$$

5. Expand:

$$(\sqrt{x} - \sqrt{2})(\sqrt{x} + \sqrt{2})$$

6. Rationalise the denominator:

$$\frac{10}{\sqrt{x} - \frac{1}{x}}$$

7. Write as a single term with a rational denominator:

$$\frac{3}{2\sqrt{x}} + \sqrt{x}$$

8. Write in simplest surd form:

a) $\sqrt{72}$

b) $\sqrt{45} + \sqrt{80}$

c) $\frac{\sqrt{48}}{\sqrt{12}}$

d) $\frac{\sqrt{18} \div \sqrt{72}}{\sqrt{8}}$

e) $\frac{4}{(\sqrt{8} \div \sqrt{2})}$

f) $\frac{16}{(\sqrt{20} \div \sqrt{12})}$

9. Expand and simplify:

a) $(2 + \sqrt{2})^2$

b) $(2 + \sqrt{2})(1 + \sqrt{8})$

c) $(1 + \sqrt{3})(1 + \sqrt{8} + \sqrt{3})$

10. Simplify, without use of a calculator:

a) $\sqrt{5}(\sqrt{45} + 2\sqrt{80})$

b) $\frac{\sqrt{98} - \sqrt{8}}{\sqrt{50}}$

11. Simplify:

$$\sqrt{98x^6} + \sqrt{128x^6}$$

12. Rationalise the denominator:

a) $\frac{\sqrt{5} + 2}{\sqrt{5}}$

b) $\frac{y - 4}{\sqrt{y} - 2}$

c) $\frac{2x - 20}{\sqrt{x} - \sqrt{10}}$

13. Evaluate without using a calculator: $\left(2 - \frac{\sqrt{7}}{2}\right)^{\frac{1}{2}} \times \left(2 + \frac{\sqrt{7}}{2}\right)^{\frac{1}{2}}$

14. Prove (without the use of a calculator):

$$\sqrt{\frac{8}{3}} + 5\sqrt{\frac{5}{3}} - \sqrt{\frac{1}{6}} = \frac{10\sqrt{15} + 3\sqrt{6}}{6}$$

15. Simplify completely by showing all your steps (do not use a calculator):

$$3^{-\frac{1}{2}} \left[\sqrt{12} + \sqrt[3]{(3\sqrt{3})} \right]$$

16. Fill in the blank surd-form number on the right hand side of the equal sign which will make the following a true statement: $-3\sqrt{6} \times -2\sqrt{24} = -\sqrt{18} \times \dots$

17. Solve for the unknown variable:

a) $3^{x-1} - 27 = 0$

b) $8^x - \frac{1}{\sqrt[3]{8}} = 0$

c) $27(4^x) = (64)3^x$

d) $\sqrt{2x-5} = 2-x$

e) $2x^{\frac{2}{3}} + 3x^{\frac{1}{3}} - 2 = 0$

18. a) Show that $\sqrt{\frac{3^{x+1} - 3^x}{3^{x-1}}} + 3$ is equal to 3

b) Hence solve $\sqrt{\frac{3^{x+1} - 3^x}{3^{x-1}}} + 3 = \left(\frac{1}{3}\right)^{x-2}$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

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2e. 225G	2f. 225H	3a. 225J	3b. 225K	3c. 225M	3d. 225N
3e. 225P	4. 225Q	5. 225R	6. 225S	7. 225T	8a. 225V
8b. 225W	8c. 225X	8d. 225Y	8e. 225Z	8f. 2262	9a. 2263
9b. 2264	9c. 2265	10a. 2266	10b. 2267	11. 2268	12a. 2269
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17a. 226J	17b. 226K	17c. 226M	17d. 226N	17e. 226P	18. 226Q



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Equations and inequalities

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2.1 Revision

EMBFF

Solving quadratic equations using factorisation

EMBFG

Terminology:	
Expression	An expression is a term or group of terms consisting of numbers, variables and the basic operators ($+$, $-$, $\times$, $\div$, x^n).
Equation	A mathematical statement that asserts that two expressions are equal.
Inequality	An inequality states the relation between two expressions ($>$, $<$, $\geq$, $\leq$).
Solution	A value or set of values that satisfy the original problem statement.
Root	A root of an equation is the value of x such that $f(x) = 0$.

A quadratic equation is an equation of the second degree; the exponent of one variable is 2.

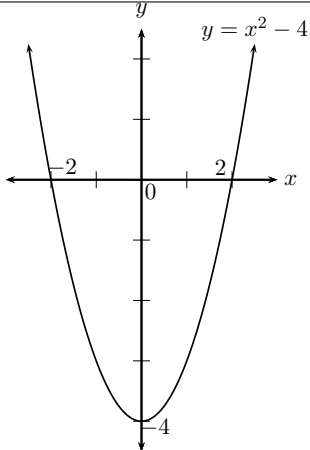
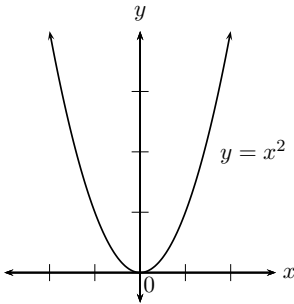
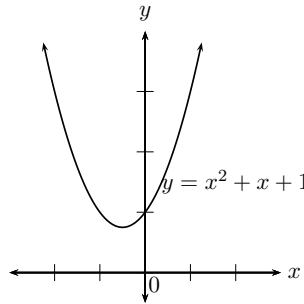
The following are examples of quadratic equations:

$$2x^2 - 5x = 12$$

$$a(a - 3) - 10 = 0$$

$$\frac{3b}{b+2} + 1 = \frac{4}{b+1}$$

A quadratic equation has at most two solutions, also referred to as roots. There are some situations, however, in which a quadratic equation has either one solution or no solutions.

		
$y = (x - 2)(x + 2)$ $= x^2 - 4$ Graph of a quadratic equation with two roots: $x = -2$ and $x = 2$.	$y = x^2$ Graph of a quadratic equation with one root: $x = 0$.	$y = x^2 + x + 1$ Graph of a quadratic equation with no real roots.

One method for solving quadratic equations is factorisation. The standard form of a quadratic equation is $ax^2 + bx + c = 0$ and it is the starting point for solving any equation by factorisation.

It is very important to note that one side of the equation must be equal to zero.

Investigation: Zero product law

Solve the following equations:

- $6 \times 0 = ?$
- $-25 \times 0 = ?$
- $0 \times 0,69 = ?$
- $7 \times ? = 0$

Now solve for the variable in each of the following:

- $6 \times m = 0$
- $32 \times x \times 2 = 0$
- $11(z - 3) = 0$
- $(k + 3)(k - 4) = 0$

To obtain the two roots we use the fact that if $a \times b = 0$, then $a = 0$ and/or $b = 0$. This is called the zero product law.

1. Rewrite the equation in the standard form $ax^2 + bx + c = 0$.
2. Divide the entire equation by any common factor of the coefficients to obtain a simpler equation of the form $ax^2 + bx + c = 0$, where a , b and c have no common factors.
3. Factorise $ax^2 + bx + c = 0$ to be of the form $(rx + s)(ux + v) = 0$.
4. The two solutions are

$$\begin{array}{ll} (rx + s) = 0 & (ux + v) = 0 \\ \text{So } x = -\frac{s}{r} & \text{So } x = -\frac{v}{u} \end{array}$$

5. Always check the solution by substituting the answer back into the original equation.

► See video: 226R at www.everythingmaths.co.za

Worked example 1: Solving quadratic equations using factorisation

QUESTION

Solve for x : $x(x - 3) = 10$

SOLUTION

Step 1: Rewrite the equation in the form $ax^2 + bx + c = 0$

Expand the brackets and subtract 10 from both sides of the equation $x^2 - 3x - 10 = 0$

Step 2: Factorise

$$(x + 2)(x - 5) = 0$$

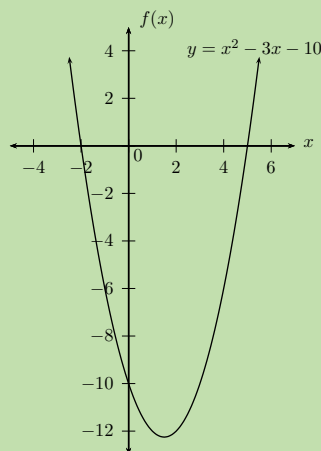
Step 3: Solve for both factors

$$\begin{array}{l} x + 2 = 0 \\ x = -2 \end{array}$$

or

$$\begin{array}{l} x - 5 = 0 \\ x = 5 \end{array}$$

The graph shows the roots of the equation $x = -2$ or $x = 5$. This graph does not form part of the answer as the question did not ask for a sketch. It is shown here for illustration purposes only.



Step 4: Check the solution by substituting both answers back into the original equation

Step 5: Write the final answer

Therefore $x = -2$ or $x = 5$.

Worked example 2: Solving quadratic equations using factorisation

QUESTION

Solve the equation: $2x^2 - 5x - 12 = 0$

SOLUTION

Step 1: There are no common factors

Step 2: The quadratic equation is already in the standard form $ax^2 + bx + c = 0$

Step 3: Factorise

We must determine the combination of factors of 2 and 12 that will give a middle term coefficient of 5.

We find that 2×1 and 3×4 give a middle term coefficient of 5 so we can factorise the equation as

$$(2x + 3)(x - 4) = 0$$

Step 4: Solve for both roots

We have

$$2x + 3 = 0$$
$$x = -\frac{3}{2}$$

or

$$x - 4 = 0$$
$$x = 4$$

Step 5: Check the solution by substituting both answers back into the original equation

Step 6: Write the final answer

Therefore, $x = -\frac{3}{2}$ or $x = 4$.

Worked example 3: Solving quadratic equations using factorisation

QUESTION

Solve for y : $y^2 - 7 = 0$

SOLUTION

Step 1: Factorise as a difference of two squares

We know that

$$(\sqrt{7})^2 = 7$$

We can write the equation as

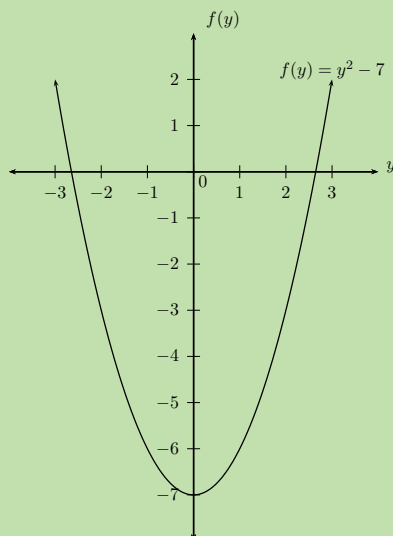
$$y^2 - (\sqrt{7})^2 = 0$$

Step 2: Factorise

$$(y - \sqrt{7})(y + \sqrt{7}) = 0$$

Therefore $y = \sqrt{7}$ or $y = -\sqrt{7}$

Even though the question did not ask for a sketch, it is often very useful to draw the graph. We can let $f(y) = y^2 - 7$ and draw a rough sketch of the graph to see where the two roots of the equation lie.



Step 3: Check the solution by substituting both answers back into the original equation

Step 4: Write the final answer

Therefore $y = \pm\sqrt{7}$.

Worked example 4: Solving quadratic equations using factorisation

QUESTION

Solve for b :

$$\frac{3b}{b+2} + 1 = \frac{4}{b+1}$$

SOLUTION

Step 1: Determine the restrictions

The restrictions are the values for b that would result in the denominator being equal to 0, which would make the fraction undefined. Therefore $b \neq -2$ and $b \neq -1$.

Step 2: Determine the lowest common denominator

The lowest common denominator is $(b+2)(b+1)$.

Step 3: Multiply each term in the equation by the lowest common denominator and simplify

$$\frac{3b(b+2)(b+1)}{b+2} + (b+2)(b+1) = \frac{4(b+2)(b+1)}{b+1}$$

$$3b(b+1) + (b+2)(b+1) = 4(b+2)$$

$$3b^2 + 3b + b^2 + 3b + 2 = 4b + 8$$

$$4b^2 + 2b - 6 = 0$$

$$2b^2 + b - 3 = 0$$

Step 4: Factorise and solve the equation

$$(2b+3)(b-1) = 0$$

$$2b+3 = 0 \text{ or } b-1 = 0$$

$$b = -\frac{3}{2} \text{ or } b = 1$$

Step 5: Check the solution by substituting both answers back into the original equation

Step 6: Write the final answer

Therefore $b = -1\frac{1}{2}$ or $b = 1$.

Worked example 5: Squaring both sides of the equation

QUESTION

Solve for m : $m + 2 = \sqrt{7 + 2m}$

SOLUTION

Step 1: Square both sides of the equation

Before we square both sides of the equation, we must make sure that the radical is the only term on one side of the equation and all other terms are on the other, otherwise squaring both sides will make the equation more complicated to solve.

$$(m+2)^2 = (\sqrt{7+2m})^2$$

Step 2: Expand the brackets and simplify

$$\begin{aligned}(m+2)^2 &= (\sqrt{7+2m})^2 \\ m^2 + 4m + 4 &= 7 + 2m \\ m^2 + 2m - 3 &= 0\end{aligned}$$

Step 3: Factorise and solve for m

$$\begin{aligned}(m-1)(m+3) &= 0 \\ \text{Therefore } m &= 1 \text{ or } m = -3\end{aligned}$$

Step 4: Check the solution by substituting both answers back into the original equation

To find the solution we squared both sides of the equation. Squaring an expression changes negative values to positives and can therefore introduce invalid answers into the solution. Therefore it is very important to check that the answers obtained are valid. To test the answers, always substitute back into the original equation.

If $m = 1$:

$$\begin{aligned}\text{RHS} &= \sqrt{7+2(1)} \\ &= \sqrt{9} \\ &= 3 \\ \text{LHS} &= 1+2 \\ &= 3 \\ \text{LHS} &= \text{RHS}\end{aligned}$$

Therefore $m = 1$ is valid.

If $m = -3$:

$$\begin{aligned}\text{RHS} &= \sqrt{7+2(-3)} \\ &= \sqrt{1} \\ &= 1 \\ \text{LHS} &= -3+2 \\ &= -1 \\ \text{LHS} &\neq \text{RHS}\end{aligned}$$

Therefore $m = -3$ is not valid.

Step 5: Write the final answer

Therefore $m = 1$.

Exercise 2 – 1: Solution by factorisation

Solve the following quadratic equations by factorisation. Answers may be left in surd form, where applicable.

1. $7t^2 + 14t = 0$
2. $12y^2 + 24y + 12 = 0$
3. $16s^2 = 400$
4. $y^2 - 5y + 6 = 0$
5. $y^2 + 5y - 36 = 0$
6. $4 + p = \sqrt{p + 6}$
7. $-y^2 - 11y - 24 = 0$
8. $13y - 42 = y^2$
9. $(x - 1)(x + 10) = -24$
10. $y^2 - 5ky + 4k^2 = 0$
11. $2y^2 - 61 = 101$
12. $2y^2 - 10 = 0$
13. $-8 + h^2 = 28$
14. $y^2 - 4 = 10$
15. $\sqrt{5 - 2p} - 4 = \frac{1}{2}p$
16. $y^2 + 28 = 100$
17. $f(2f + 1) = 15$
18. $2x = \sqrt{21x - 5}$
19. $\frac{5y}{y - 2} + \frac{3}{y} + 2 = \frac{-6}{y^2 - 2y}$
20. $\frac{x + 9}{x^2 - 9} + \frac{1}{x + 3} = \frac{2}{x - 3}$
21. $\frac{y - 2}{y + 1} = \frac{2y + 1}{y - 7}$
22. $1 + \frac{t - 2}{t - 1} = \frac{5}{t^2 - 4t + 3} + \frac{10}{3 - t}$
23. $\frac{4}{m + 3} + \frac{4}{4 - m^2} = \frac{5m - 5}{m^2 + m - 6}$
24. $5\sqrt{5t + 1} - 4 = 5t + 1$

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2.2 Completing the square

EMBFJ

Investigation: Completing the square

Can you solve each equation using two different methods?

1. $x^2 - 4 = 0$
2. $x^2 - 8 = 0$
3. $x^2 - 4x + 4 = 0$

$$4. \ x^2 - 4x - 4 = 0$$

Factorising the last equation is quite difficult. Use the previous examples as a hint and try to create a difference of two squares.

▶ See video: 227P at www.everythingmaths.co.za

We have seen that expressions of the form $x^2 - b^2$ are known as differences of squares and can be factorised as $(x - b)(x + b)$. This simple factorisation leads to another technique for solving quadratic equations known as completing the square.

Consider the equation $x^2 - 2x - 1 = 0$. We cannot easily factorise this expression. When we expand the perfect square $(x - 1)^2$ and examine the terms we see that $(x - 1)^2 = x^2 - 2x + 1$.

We compare the two equations and notice that only the constant terms are different. We can create a perfect square by adding and subtracting the same amount to the original equation.

$$\begin{aligned} x^2 - 2x - 1 &= 0 \\ (x^2 - 2x + 1) - 1 - 1 &= 0 \\ (x^2 - 2x + 1) - 2 &= 0 \\ (x - 1)^2 - 2 &= 0 \end{aligned}$$

Method 1: Take square roots on both sides of the equation to solve for x .

$$\begin{aligned} (x - 1)^2 - 2 &= 0 \\ (x - 1)^2 &= 2 \\ \sqrt{(x - 1)^2} &= \pm\sqrt{2} \\ x - 1 &= \pm\sqrt{2} \\ x &= 1 \pm \sqrt{2} \\ \text{Therefore } x &= 1 + \sqrt{2} \text{ or } x = 1 - \sqrt{2} \end{aligned}$$

Very important: Always remember to include both a positive and a negative answer when taking the square root, since $2^2 = 4$ and $(-2)^2 = 4$.

Method 2: Factorise the expression as a difference of two squares using $2 = (\sqrt{2})^2$.

We can write

$$\begin{aligned} (x - 1)^2 - 2 &= 0 \\ (x - 1)^2 - (\sqrt{2})^2 &= 0 \\ ((x - 1) + \sqrt{2})((x - 1) - \sqrt{2}) &= 0 \end{aligned}$$

The solution is then

$$\begin{aligned}(x - 1) + \sqrt{2} &= 0 \\ x &= 1 - \sqrt{2}\end{aligned}$$

or

$$\begin{aligned}(x - 1) - \sqrt{2} &= 0 \\ x &= 1 + \sqrt{2}\end{aligned}$$

Method for solving quadratic equations by completing the square

1. Write the equation in the standard form $ax^2 + bx + c = 0$.
2. Make the coefficient of the x^2 term equal to 1 by dividing the entire equation by a .
3. Take half the coefficient of the x term and square it; then add and subtract it from the equation so that the equation remains mathematically correct. In the example above, we added 1 to complete the square and then subtracted 1 so that the equation remained true.
4. Write the left hand side as a difference of two squares.
5. Factorise the equation in terms of a difference of squares and solve for x .

📺 See video: [227Q](#) at www.everythingmaths.co.za

Worked example 6: Solving quadratic equations by completing the square

QUESTION

Solve by completing the square: $x^2 - 10x - 11 = 0$

SOLUTION

Step 1: The equation is already in the form $ax^2 + bx + c = 0$

Step 2: Make sure the coefficient of the x^2 term is equal to 1

$$x^2 - 10x - 11 = 0$$

Step 3: Take half the coefficient of the x term and square it; then add and subtract it from the equation

The coefficient of the x term is -10 . Half of the coefficient of the x term is -5 and the square of it is 25. Therefore $x^2 - 10x + 25 - 25 - 11 = 0$.

Step 4: Write the trinomial as a perfect square

$$\begin{aligned}(x^2 - 10x + 25) - 25 - 11 &= 0 \\ (x - 5)^2 - 36 &= 0\end{aligned}$$

Step 5: Method 1: Take square roots on both sides of the equation

$$\begin{aligned}(x - 5)^2 - 36 &= 0 \\ (x - 5)^2 &= 36 \\ x - 5 &= \pm\sqrt{36}\end{aligned}$$

Important: When taking a square root always remember that there is a positive and negative answer, since $(6)^2 = 36$ and $(-6)^2 = 36$.

$$x - 5 = \pm 6$$

Step 6: Solve for x

$$x = -1 \text{ or } x = 11$$

Step 7: Method 2: Factorise equation as a difference of two squares

$$\begin{aligned}(x - 5)^2 - (6)^2 &= 0 \\ [(x - 5) + 6][(x - 5) - 6] &= 0\end{aligned}$$

Step 8: Simplify and solve for x

$$\begin{aligned}(x + 1)(x - 11) &= 0 \\ \therefore x &= -1 \text{ or } x = 11\end{aligned}$$

Step 9: Write the final answer

$$x = -1 \text{ or } x = 11$$

Notice that both methods produce the same answer. These roots are rational because 36 is a perfect square.

Worked example 7: Solving quadratic equations by completing the square

QUESTION

Solve by completing the square: $2x^2 - 6x - 10 = 0$

SOLUTION

Step 1: The equation is already in standard form $ax^2 + bx + c = 0$

Step 2: Make sure that the coefficient of the x^2 term is equal to 1

The coefficient of the x^2 term is 2. Therefore divide the entire equation by 2:

$$x^2 - 3x - 5 = 0$$

Step 3: Take half the coefficient of the x term, square it; then add and subtract it from the equation

The coefficient of the x term is -3 , so then $\left(\frac{-3}{2}\right)^2 = \frac{9}{4}$:

$$\left(x^2 - 3x + \frac{9}{4}\right) - \frac{9}{4} - 5 = 0$$

Step 4: Write the trinomial as a perfect square

$$\left(x - \frac{3}{2}\right)^2 - \frac{9}{4} - \frac{20}{4} = 0$$

$$\left(x - \frac{3}{2}\right)^2 - \frac{29}{4} = 0$$

Step 5: Method 1: Take square roots on both sides of the equation

$$\left(x - \frac{3}{2}\right)^2 - \frac{29}{4} = 0$$

$$\left(x - \frac{3}{2}\right)^2 = \frac{29}{4}$$

$$x - \frac{3}{2} = \pm\sqrt{\frac{29}{4}}$$

Remember: When taking a square root there is a positive and a negative answer.

Step 6: Solve for x

$$\begin{aligned}
 x - \frac{3}{2} &= \pm \sqrt{\frac{29}{4}} \\
 x &= \frac{3}{2} \pm \frac{\sqrt{29}}{2} \\
 &= \frac{3 \pm \sqrt{29}}{2}
 \end{aligned}$$

Step 7: Method 2: Factorise equation as a difference of two squares

$$\begin{aligned}
 \left(x - \frac{3}{2}\right)^2 - \frac{29}{4} &= 0 \\
 \left(x - \frac{3}{2}\right)^2 - \left(\sqrt{\frac{29}{4}}\right)^2 &= 0 \\
 \left(x - \frac{3}{2} - \sqrt{\frac{29}{4}}\right) \left(x - \frac{3}{2} + \sqrt{\frac{29}{4}}\right) &= 0
 \end{aligned}$$

Step 8: Solve for x

$$\begin{aligned}
 \left(x - \frac{3}{2} - \frac{\sqrt{29}}{2}\right) \left(x - \frac{3}{2} + \frac{\sqrt{29}}{2}\right) &= 0 \\
 \text{Therefore } x &= \frac{3}{2} + \frac{\sqrt{29}}{2} \text{ or } x = \frac{3}{2} - \frac{\sqrt{29}}{2}
 \end{aligned}$$

Notice that these roots are irrational since 29 is not a perfect square.

► See video: [227R](https://www.everythingmaths.co.za/227R) at www.everythingmaths.co.za

Exercise 2 – 2: Solution by completing the square

1. Solve the following equations by completing the square:

a) $x^2 + 10x - 2 = 0$

b) $x^2 + 4x + 3 = 0$

c) $p^2 - 5 = -8p$

d) $2(6x + x^2) = -4$

e) $x^2 + 5x + 9 = 0$

f) $t^2 + 30 = 2(10 - 8t)$

g) $3x^2 + 6x - 2 = 0$

h) $z^2 + 8z - 6 = 0$

i) $2z^2 = 11z$

j) $5 + 4z - z^2 = 0$

2. Solve for k in terms of a : $k^2 + 6k + a = 0$
3. Solve for y in terms of p , q and r : $py^2 + qy + r = 0$

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2.3 Quadratic formula

EMBFK

It is not always possible to solve a quadratic equation by factorisation and it can take a long time to complete the square. The method of completing the square provides a way to derive a formula that can be used to solve any quadratic equation. The quadratic formula provides an easy and fast way to solve quadratic equations.

Consider the standard form of the quadratic equation $ax^2 + bx + c = 0$. Divide both sides by a ($a \neq 0$) to get

$$x^2 + \frac{bx}{a} + \frac{c}{a} = 0$$

Now using the method of completing the square, we must halve the coefficient of x and square it. We then add and subtract $\left(\frac{b}{2a}\right)^2$ so that the equation remains true.

$$\begin{aligned} x^2 + \frac{bx}{a} + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a} &= 0 \\ \left(x^2 + \frac{bx}{a} + \frac{b^2}{4a^2}\right) - \frac{b^2}{4a^2} + \frac{c}{a} &= 0 \\ \left(x + \frac{b}{2a}\right)^2 - \frac{b^2 - 4ac}{4a^2} &= 0 \end{aligned}$$

We add the constant to both sides and take the square root of both sides of the equation, being careful to include a positive and negative answer.

$$\begin{aligned}\left(x + \frac{b}{2a}\right)^2 &= \frac{b^2 - 4ac}{4a^2} \\ \sqrt{\left(x + \frac{b}{2a}\right)^2} &= \pm \sqrt{\frac{b^2 - 4ac}{4a^2}} \\ x + \frac{b}{2a} &= \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ x &= -\frac{b}{2a} \pm \frac{\sqrt{b^2 - 4ac}}{2a} \\ x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}\end{aligned}$$

Therefore, for any quadratic equation $ax^2 + bx + c = 0$ we can determine two roots

$$x = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \text{ or } x = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

It is important to notice that the expression $b^2 - 4ac$ must be greater than or equal to zero for the roots of the quadratic to be real. If the expression under the square root sign is less than zero, then the roots are non-real (imaginary).

► See video: 2287 at www.everythingmaths.co.za

Worked example 8: Using the quadratic formula

QUESTION

Solve for x and leave your answer in simplest surd form: $2x^2 + 3x = 7$

SOLUTION

Step 1: Check whether the expression can be factorised

The expression cannot be factorised, so the general quadratic formula must be used.

Step 2: Write the equation in the standard form $ax^2 + bx + c = 0$

$$2x^2 + 3x - 7 = 0$$

Step 3: Identify the coefficients to substitute into the formula

$$a = 2; \quad b = 3; \quad c = -7$$

Step 4: Apply the quadratic formula

Always write down the formula first and then substitute the values of a , b and c .

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{-(3) \pm \sqrt{(3)^2 - 4(2)(-7)}}{2(2)} \\
 &= \frac{-3 \pm \sqrt{65}}{4}
 \end{aligned}$$

Step 5: Write the final answer

The two roots are $x = \frac{-3 + \sqrt{65}}{4}$ or $x = \frac{-3 - \sqrt{65}}{4}$.

Worked example 9: Using the quadratic formula**QUESTION**

Find the roots of the function $f(x) = x^2 - 5x + 8$.

SOLUTION**Step 1: Finding the roots**

To determine the roots of $f(x)$, we let $x^2 - 5x + 8 = 0$.

Step 2: Check whether the expression can be factorised

The expression cannot be factorised, so the general quadratic formula must be used.

Step 3: Identify the coefficients to substitute into the formula

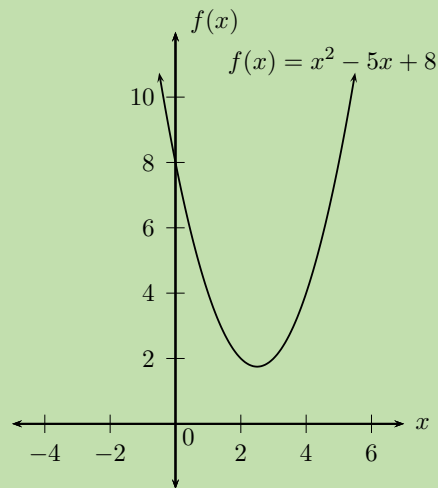
$$a = 1; \quad b = -5; \quad c = 8$$

Step 4: Apply the quadratic formula

$$\begin{aligned}
 x &= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \\
 &= \frac{-(-5) \pm \sqrt{(-5)^2 - 4(1)(8)}}{2(1)} \\
 &= \frac{5 \pm \sqrt{-7}}{2}
 \end{aligned}$$

Step 5: Write the final answer

There are no real roots for $f(x) = x^2 - 5x + 8$ since the expression under the square root is negative ($\sqrt{-7}$ is not a real number). This means that the graph of the quadratic function has no x -intercepts; the entire graph lies above the x -axis.



📺 See video: 2288 at www.everythingmaths.co.za

Exercise 2 – 3: Solution by the quadratic formula

Solve the following using the quadratic formula.

- | | |
|-------------------------|------------------------|
| 1. $3t^2 + t - 4 = 0$ | 6. $5t^2 + 3t - 3 = 0$ |
| 2. $x^2 - 5x - 3 = 0$ | 7. $t^2 - 4t + 2 = 0$ |
| 3. $2t^2 + 6t + 5 = 0$ | 8. $9(k^2 - 1) = 7k$ |
| 4. $2p(2p + 1) = 2$ | 9. $3f - 2 = -2f^2$ |
| 5. $-3t^2 + 5t - 8 = 0$ | 10. $t^2 + t + 1 = 0$ |

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It is often useful to make a substitution for a repeated expression in a quadratic equation. This makes the equation simpler and much easier to solve.

Worked example 10: Solving by substitution

QUESTION

Solve for x : $x^2 - 2x - \frac{3}{x^2 - 2x} = 2$

SOLUTION

Step 1: Determine the restrictions for x

The restrictions are the values for x that would result in the denominator being equal to 0, which would make the fraction undefined. Therefore $x \neq 0$ and $x \neq 2$.

Step 2: Substitute a single variable for the repeated expression

We notice that $x^2 - 2x$ is a repeated expression and we therefore let $k = x^2 - 2x$ so that the equation becomes

$$k - \frac{3}{k} = 2$$

Step 3: Determine the restrictions for k

The restrictions are the values for k that would result in the denominator being equal to 0, which would make the fraction undefined. Therefore $k \neq 0$.

Step 4: Solve for k

$$k - \frac{3}{k} = 2$$

$$k^2 - 3 = 2k$$

$$k^2 - 2k - 3 = 0$$

$$(k + 1)(k - 3) = 0$$

$$\text{Therefore } k = -1 \text{ or } k = 3$$

We check these two roots against the restrictions for k and confirm that both are valid.

Step 5: Use values obtained for k to solve for the original variable x

For $k = -1$

$$x^2 - 2x = -1$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)(x - 1) = 0$$

Therefore $x = 1$

For $k = 3$

$$x^2 - 2x = 3$$

$$x^2 - 2x - 3 = 0$$

$$(x + 1)(x - 3) = 0$$

Therefore $x = -1$ or $x = 3$

We check these roots against the restrictions for x and confirm that all three values are valid.

Step 6: Write the final answer

The roots of the equation are $x = -1$, $x = 1$ and $x = 3$.

Exercise 2 – 4:

Solve the following quadratic equations by substitution:

1. $-24 = 10(x^2 + 5x) + (x^2 + 5x)^2$

4. $x^2 - 18 + x + \frac{72}{x^2 + x} = 0$

2. $(x^2 - 2x)^2 - 8 = 7(x^2 - 2x)$

5. $x^2 - 4x + 10 - 7(4x - x^2) = -2$

3. $x^2 + 3x - \frac{56}{x(x + 3)} = 26$

6. $\frac{9}{x^2 + 2x - 12} = x^2 + 2x - 12$

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We have seen that the roots are the solutions obtained from solving a quadratic equation. Given the roots, we are also able to work backwards to determine the original quadratic equation.

Worked example 11: Finding an equation when the roots are given**QUESTION**

Find an equation with roots 13 and -5 .

SOLUTION

Step 1: Assign a variable and write roots as two equations

$$x = 13 \text{ or } x = -5$$

Use additive inverses to get zero on the right-hand sides

$$x - 13 = 0 \text{ or } x + 5 = 0$$

Step 2: Write down as the product of two factors

$$(x - 13)(x + 5) = 0$$

Notice that the signs in the brackets are opposite of the given roots.

Step 3: Expand the brackets

$$x^2 - 8x - 65 = 0$$

Note that if each term in the equation is multiplied by a constant then there could be other possible equations which would have the same roots. For example,

Multiply by 2:

$$2x^2 - 16x - 130 = 0$$

Multiply by -3 :

$$-3x^2 + 24x + 195 = 0$$

Worked example 12: Finding an equation when the roots are fractions

QUESTION

Find an equation with roots $-\frac{3}{2}$ and 4.

SOLUTION

Step 1: Assign a variable and write roots as two equations

$$x = 4 \text{ or } x = -\frac{3}{2}$$

Use additive inverses to get zero on the right-hand sides.

$$x - 4 = 0 \text{ or } x + \frac{3}{2} = 0$$

Multiply the second equation through by 2 to remove the fraction.

$$x - 4 = 0 \text{ or } 2x + 3 = 0$$

Step 2: Write down as the product of two factors

$$(2x + 3)(x - 4) = 0$$

Step 3: Expand the brackets

The quadratic equation is $2x^2 - 5x - 12 = 0$.

Exercise 2 – 5: Finding the equation

1. Determine a quadratic equation for a graph that has roots 3 and -2 .
2. Find a quadratic equation for a graph that has x -intercepts of $(-4; 0)$ and $(4; 0)$.
3. Determine a quadratic equation of the form $ax^2 + bx + c = 0$, where a, b and c are integers, that has roots $-\frac{1}{2}$ and 3.
4. Determine the value of k and the other root of the quadratic equation $kx^2 - 7x + 4 = 0$ given that one of the roots is $x = 1$.
5. One root of the equation $2x^2 - 3x = p$ is $2\frac{1}{2}$. Find p and the other root.

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Exercise 2 – 6: Mixed exercises

Solve the following quadratic equations by either factorisation, using the quadratic formula or completing the square:

- Always try to factorise first, then use the formula if the trinomial cannot be factorised.
- In a test or examination, only use the method of completing the square when specifically asked.
- Answers can be left in surd or decimal form.

1. $24y^2 + 61y - 8 = 0$

10. $6y^2 + 7y - 24 = 0$

2. $8x^2 + 16x = 42$

11. $3 = x(2x - 5)$

3. $9t^2 = 24t - 12$

12. $-18y^2 - 55y - 25 = 0$

4. $-5y^2 + 0y + 5 = 0$

13. $-25y^2 + 25y - 4 = 0$

5. $3m^2 + 12 = 15m$

14. $8(1 - 4g^2) + 24g = 0$

6. $49y^2 + 0y - 25 = 0$

15. $9y^2 - 13y - 10 = 0$

7. $72 = 66w - 12w^2$

16. $(7p - 3)(5p + 1) = 0$

8. $-40y^2 + 58y - 12 = 0$

17. $-81y^2 - 99y - 18 = 0$

9. $37n + 72 - 24n^2 = 0$

18. $14y^2 - 81y + 81 = 0$

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- | | | | | | |
|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
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| 7. 2298 | 8. 2299 | 9. 229B | 10. 229C | 11. 229D | 12. 229F |
| 13. 229G | 14. 229H | 15. 229J | 16. 229K | 17. 229M | 18. 229N |



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2.6 Nature of roots

EMBFP

Investigation:

1. Use the quadratic formula to determine the roots of the quadratic equations given below and take special note of:

- the expression under the square root sign and
- the type of number for the final answer (rational/irrational/real/imaginary)

a) $x^2 - 6x + 9 = 0$

b) $x^2 - 4x + 3 = 0$

c) $x^2 - 4x - 3 = 0$

d) $x^2 - 4x + 7 = 0$

2. Choose the appropriate words from the table to describe the roots obtained for the equations above.

rational	unequal	real
imaginary	not perfect square	equal
perfect square	irrational	undefined

3. The expression under the square root, $b^2 - 4ac$, is called the discriminant. Can you make a conjecture about the relationship between the discriminant and the roots of quadratic equations?

The discriminant

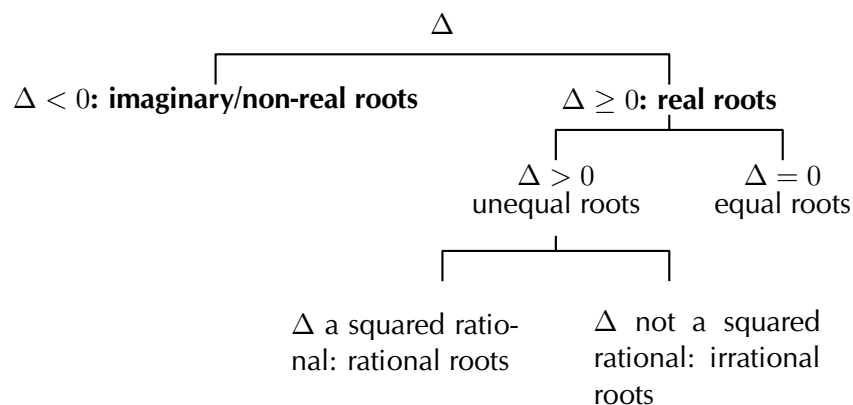
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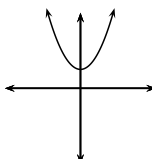
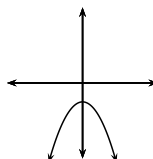
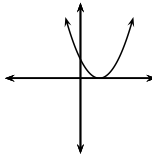
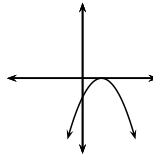
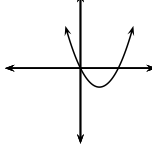
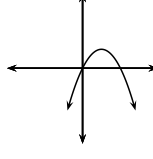
The discriminant is defined as $\Delta = b^2 - 4ac$

This is the expression under the square root in the quadratic formula. The discriminant determines the nature of the roots of a quadratic equation. The word 'nature' refers to the types of numbers the roots can be — namely real, rational, irrational or imaginary. Δ is the Greek symbol for the letter D.

For a quadratic function $f(x) = ax^2 + bx + c$, the solutions to the equation $f(x) = 0$ are given by the formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{-b \pm \sqrt{\Delta}}{2a}$

- If $\Delta < 0$, then roots are imaginary (non-real) and beyond the scope of this book.
- If $\Delta \geq 0$, the expression under the square root is non-negative and therefore roots are real. For real roots, we have the following further possibilities.
- If $\Delta = 0$, the roots are equal and we can say that there is only one root.
- If $\Delta > 0$, the roots are unequal and there are two further possibilities.
- Δ is the square of a rational number: the roots are rational.
- Δ is not the square of a rational number: the roots are irrational and can be expressed in decimal or surd form.



Nature of roots	Discriminant	$a > 0$	$a < 0$
Roots are non-real	$\Delta < 0$		
Roots are real and equal	$\Delta = 0$		
Roots are real and unequal: - rational roots - irrational roots	$\Delta > 0$ $\Delta = \text{squared rational}$ $\Delta = \text{not squared rational}$		

▶ See video: 229P at www.everythingmaths.co.za

Worked example 13: Nature of roots

QUESTION

Show that the roots of $x^2 - 2x - 7 = 0$ are irrational.

SOLUTION

Step 1: Interpret the question

For roots to be real and irrational, we need to calculate Δ and show that it is greater than zero and not a perfect square.

Step 2: Check that the equation is in standard form $ax^2 + bx + c = 0$

$$x^2 - 2x - 7 = 0$$

Step 3: Identify the coefficients to substitute into the formula for the discriminant

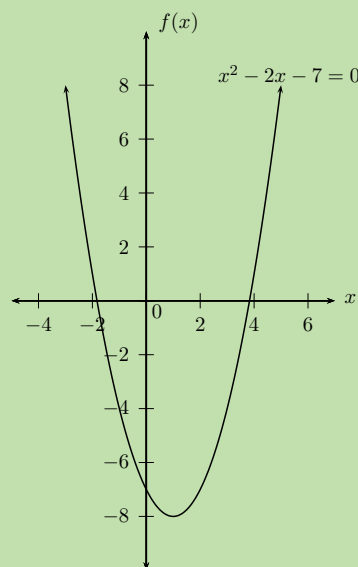
$$a = 1; \quad b = -2; \quad c = -7$$

Step 4: Write down the formula and substitute values

$$\begin{aligned} \Delta &= b^2 - 4ac \\ &= (-2)^2 - 4(1)(-7) \\ &= 4 + 28 \\ &= 32 \end{aligned}$$

We know that $32 > 0$ and is not a perfect square.

The graph below shows the roots of the equation $x^2 - 2x - 7 = 0$. Note that the graph does not form part of the answer and is included for illustration purposes only.



Step 5: Write the final answer

We have calculated that $\Delta > 0$ and is not a perfect square, therefore we can conclude that the roots are **real, unequal** and **irrational**.

Worked example 14: Nature of roots

QUESTION

For which value(s) of k will the roots of $6x^2 + 6 = 4kx$ be real and equal?

SOLUTION

Step 1: Interpret the question

For roots to be real and equal, we need to solve for the value(s) of k such that $\Delta = 0$.

Step 2: Check that the equation is in standard form $ax^2 + bx + c = 0$

$$6x^2 - 4kx + 6 = 0$$

Step 3: Identify the coefficients to substitute into the formula for the discriminant

$$a = 6; \quad b = -4k; \quad c = 6$$

Step 4: Write down the formula and substitute values

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (-4k)^2 - 4(6)(6) \\ &= 16k^2 - 144\end{aligned}$$

For roots to be real and equal, $\Delta = 0$.

$$\begin{aligned}\Delta &= 0 \\ 16k^2 - 144 &= 0 \\ 16(k^2 - 9) &= 0 \\ (k - 3)(k + 3) &= 0\end{aligned}$$

Therefore $k = 3$ or $k = -3$.

Step 5: Check both answers by substituting back into the original equation

For $k = 3$:

$$6x^2 - 4(3)x + 6 = 0$$

$$6x^2 - 12x + 6 = 0$$

$$x^2 - 2x + 1 = 0$$

$$(x - 1)(x - 1) = 0$$

$$(x - 1)^2 = 0$$

$$\text{Therefore } x = 1$$

We see that for $k = 3$ the quadratic equation has real, equal roots $x = 1$.

For $k = -3$:

$$6x^2 - 4(-3)x + 6 = 0$$

$$6x^2 + 12x + 6 = 0$$

$$x^2 + 2x + 1 = 0$$

$$(x + 1)(x + 1) = 0$$

$$(x + 1)^2 = 0$$

$$\text{Therefore } x = -1$$

We see that for $k = -3$ the quadratic equation has real, equal roots $x = -1$.

Step 6: Write the final answer

For the roots of the quadratic equation to be **real** and **equal**, $k = 3$ or $k = -3$.

Worked example 15: Nature of roots

QUESTION

Show that the roots of $(x + h)(x + k) = 4d^2$ are real for all real values of h, k and d .

SOLUTION

Step 1: Interpret the question

For roots to be real, we need to calculate Δ and show that $\Delta \geq 0$ for all real values of h, k and d .

Step 2: Check that the equation is in standard form $ax^2 + bx + c = 0$

Expand the brackets and gather like terms

$$\begin{aligned}(x + h)(x + k) &= 4d^2 \\ x^2 + hx + kx + hk - 4d^2 &= 0 \\ x^2 + (h + k)x + (hk - 4d^2) &= 0\end{aligned}$$

Step 3: Identify the coefficients to substitute into the formula for the discriminant

$$a = 1; \quad b = h + k; \quad c = hk - 4d^2$$

Step 4: Write down the formula and substitute values

$$\begin{aligned}\Delta &= b^2 - 4ac \\ &= (h + k)^2 - 4(1)(hk - 4d^2) \\ &= h^2 + 2hk + k^2 - 4hk + 16d^2 \\ &= h^2 - 2hk + k^2 + 16d^2 \\ &= (h - k)^2 + (4d)^2\end{aligned}$$

For roots to be real, $\Delta \geq 0$.

$$\begin{aligned}\text{We know that } (4d)^2 &\geq 0 \\ \text{and } (h - k)^2 &\geq 0 \\ \text{so then } (h - k)^2 + (4d)^2 &\geq 0 \\ \text{therefore } \Delta &\geq 0\end{aligned}$$

Step 5: Write the final answer

We have shown that $\Delta \geq 0$, therefore the roots are **real** for all real values of h, k and d .

Exercise 2 – 7: From past papers

1. Determine the nature of the roots for each of the following equations:

a) $x^2 + 3x = -2$

f) $0 = p^2 + 5p + 8$

b) $x^2 + 9 = 6x$

g) $x^2 = 36$

c) $6y^2 - 6y - 1 = 0$

h) $4m + m^2 = 1$

d) $4t^2 - 19t - 5 = 0$

i) $11 - 3x + x^2 = 0$

e) $z^2 = 3$

j) $y^2 + \frac{1}{4} = y$

2. Given: $x^2 + bx - 2 + k(x^2 + 3x + 2) = 0$, ($k \neq -1$)

a) Show that the discriminant is given by: $\Delta = k^2 + 6bk + b^2 + 8$

- b) If $b = 0$, discuss the nature of the roots of the equation.
 c) If $b = 2$, find the value(s) of k for which the roots are equal.

[IEB, Nov. 2001, HG]

3. Show that $k^2x^2 + 2 = kx - x^2$ has non-real roots for all real values for k .

[IEB, Nov. 2002, HG]

4. The equation $x^2 + 12x = 3kx^2 + 2$ has real roots.

- a) Find the greatest value of value k such that $k \in \mathbb{Z}$.
 b) Find one rational value of k for which the above equation has rational roots.

[IEB, Nov. 2003, HG]

5. Consider the equation:

$$k = \frac{x^2 - 4}{2x - 5}$$

where $x \neq \frac{5}{2}$.

- a) Find a value of k for which the roots are equal.
 b) Find an integer k for which the roots of the equation will be rational and unequal.

[IEB, Nov. 2004, HG]

6. a) Prove that the roots of the equation $x^2 - (a + b)x + ab - p^2 = 0$ are real for all real values of a , b and p .
 b) When will the roots of the equation be equal?

[IEB, Nov. 2005, HG]

7. If b and c can take on only the values 1, 2 or 3, determine all pairs $(b; c)$ such that $x^2 + bx + c = 0$ has real roots.

[IEB, Nov. 2005, HG]

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 4. [22B5](#) 5. [22B6](#) 6. [22B7](#) 7. [22B8](#)



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Quadratic inequalities can be of the following forms:

$$ax^2 + bx + c > 0$$

$$ax^2 + bx + c \geq 0$$

$$ax^2 + bx + c < 0$$

$$ax^2 + bx + c \leq 0$$

To solve a quadratic inequality we must determine which part of the graph of a quadratic function lies above or below the x -axis. An inequality can therefore be solved graphically using a graph or algebraically using a table of signs to determine where the function is positive and negative.

Worked example 16: Solving quadratic inequalities

QUESTION

Solve for x : $x^2 - 5x + 6 \geq 0$

SOLUTION

Step 1: Factorise the quadratic

$$(x - 3)(x - 2) \geq 0$$

Step 2: Determine the critical values of x

From the factorised quadratic we see that the values for which the inequality is equal to zero are $x = 3$ and $x = 2$. These are called the critical values of the inequality and they are used to complete a table of signs.

Step 3: Complete a table of signs

We must determine where each factor of the inequality is positive and negative on the number line:

- to the left (in the negative direction) of the critical value
- equal to the critical value
- to the right (in the positive direction) of the critical value

In the final row of the table we determine where the inequality is positive and negative by finding the product of the factors and their respective signs.

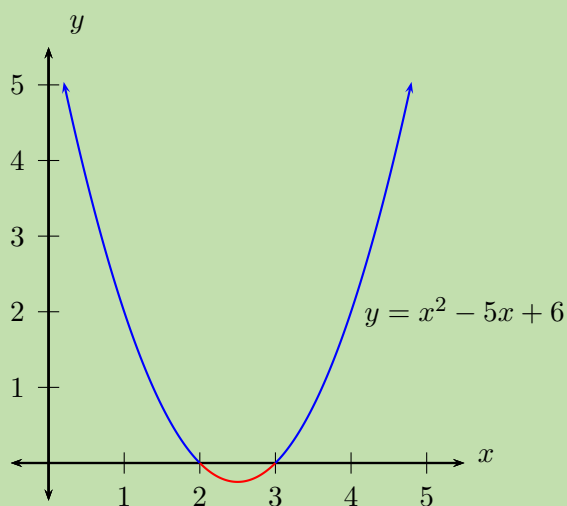
Critical values		$x = 2$		$x = 3$	
$x - 3$	–	–	–	0	+
$x - 2$	–	0	+	+	+
$f(x) = (x - 3)(x - 2)$	+	0	–	0	+

From the table we see that $f(x)$ is greater than or equal to zero for $x \leq 2$ or $x \geq 3$.

Step 4: A rough sketch of the graph

The graph below does not form part of the answer and is included for illustration purposes only. A graph of the quadratic helps us determine the answer to the inequality. We can find the answer graphically by seeing where the graph lies above or below the x -axis.

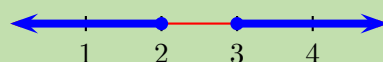
- From the standard form, $x^2 - 5x + 6$, $a > 0$ and therefore the graph is a “smile” and has a minimum turning point.
- From the factorised form, $(x - 3)(x - 2)$, we know the x -intercepts are $(2; 0)$ and $(3; 0)$.



The graph is above or on the x -axis for $x \leq 2$ or $x \geq 3$.

Step 5: Write the final answer and represent on a number line

$$x^2 - 5x + 6 \geq 0 \text{ for } x \leq 2 \text{ or } x \geq 3$$



QUESTION

Solve for x : $4x^2 - 4x + 1 \leq 0$

SOLUTION

Step 1: Factorise the quadratic

$$\begin{aligned}(2x - 1)(2x - 1) &\leq 0 \\ (2x - 1)^2 &\leq 0\end{aligned}$$

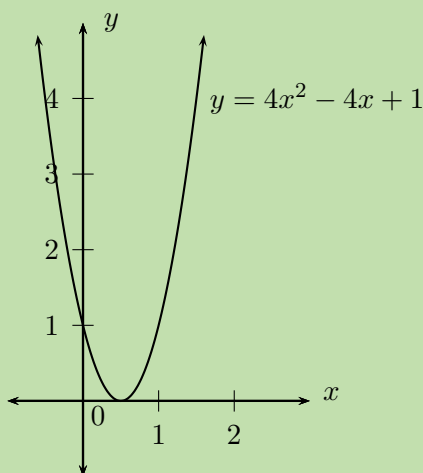
Step 2: Determine the critical values of x

From the factorised quadratic we see that the value for which the inequality is equal to zero is $x = \frac{1}{2}$. We know that $a^2 > 0$ for any real number a , $a \neq 0$, so then $(2x - 1)^2$ will never be negative.

Step 3: A rough sketch of the graph

The graph below does not form part of the answer and is included for illustration purposes only.

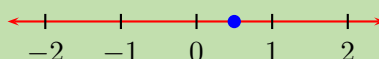
- From the standard form, $4x^2 - 4x + 1$, $a > 0$ and therefore the graph is a “smile” and has a minimum turning point.
- From the factorised form, $(2x - 1)(2x - 1)$, we know there is only one x -intercept at $(\frac{1}{2}; 0)$.



Notice that no part of the graph lies below the x -axis.

Step 4: Write the final answer and represent on a number line

$$4x^2 - 4x + 1 \leq 0 \text{ for } x = \frac{1}{2}$$



Worked example 18: Solving quadratic inequalities

QUESTION

Solve for x : $-x^2 - 3x + 5 > 0$

SOLUTION

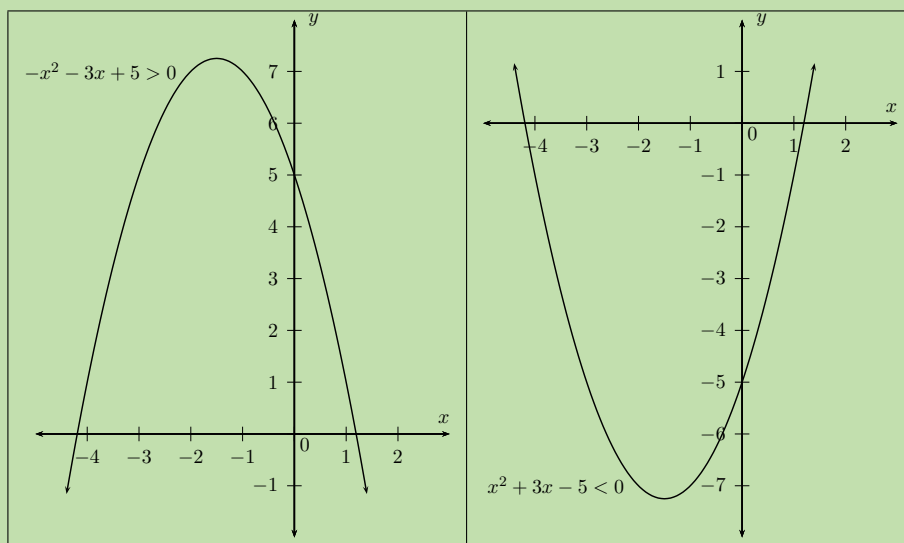
Step 1: Examine the form of the inequality

Notice that the coefficient of the x^2 term is -1 . **Remember** that if we multiply or divide an inequality by a negative number, then the inequality sign changes direction. So we can write the same inequality in different ways and still get the same answer, as shown below.

$$-x^2 - 3x + 5 > 0$$

Multiply by -1 and change direction of the inequality sign

$$x^2 + 3x - 5 < 0$$



From this rough sketch, we can see that both inequalities give the same solution; the values of x that lie between the two x -intercepts.

Step 2: Factorise the quadratic

We notice that $-x^2 - 3x + 5 > 0$ cannot be easily factorised. So we let $-x^2 - 3x + 5 = 0$ and use the quadratic formula to determine the roots of the equation.

$$-x^2 - 3x + 5 = 0$$

$$x^2 + 3x - 5 = 0$$

$$\begin{aligned}\therefore x &= \frac{-3 \pm \sqrt{(3)^2 - 4(1)(-5)}}{2(1)} \\ &= \frac{-3 \pm \sqrt{29}}{2} \\ x_1 &= \frac{-3 - \sqrt{29}}{2} \approx -4,2 \\ x_2 &= \frac{-3 + \sqrt{29}}{2} \approx 1,2\end{aligned}$$

Therefore we can write, correct to one decimal place,

$$\begin{aligned}x^2 + 3x - 5 &< 0 \\ \text{as } (x - 1,2)(x + 4,2) &< 0\end{aligned}$$

Step 3: Determine the critical values of x

From the factorised quadratic we see that the critical values are $x = 1,2$ and $x = -4,2$.

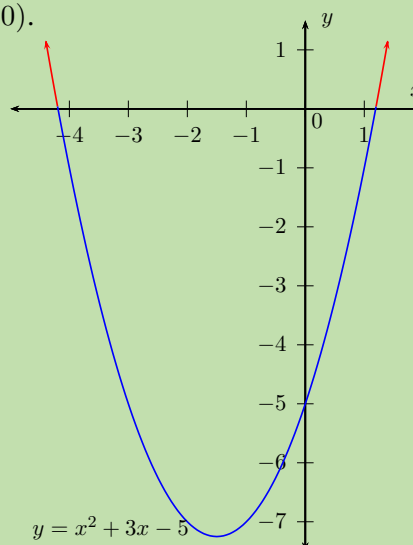
Step 4: Complete a table of signs

Critical values		$x = -4,2$		$x = 1,2$	
$x + 4,2$	–	0	+	+	+
$x - 1,2$	–	–	–	0	+
$f(x) = (x + 4,2)(x - 1,2)$	+	0	–	0	+

From the table we see that the function is negative for $-4,2 < x < 1,2$.

Step 5: A sketch of the graph

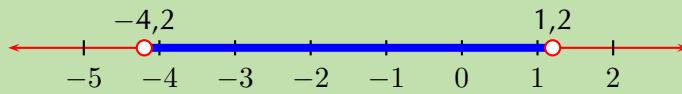
- From the standard form, $x^2 + 3x - 5$, $a > 0$ and therefore the graph is a “smile” and has a minimum turning point.
- From the factorised form, $(x - 1,2)(x + 4,2)$, we know the x -intercepts are $(-4,2; 0)$ and $(1,2; 0)$.



From the graph we see that the function lies below the x -axis between $-4,2$ and $1,2$.

Step 6: Write the final answer and represent on a number line

$$x^2 + 3x - 5 < 0 \text{ for } -4,2 < x < 1,2$$



Important: When working with an inequality in which the variable is in the denominator, a different approach is needed. Always remember to check for restrictions.

Worked example 19: Solving quadratic inequalities with fractions

QUESTION

Solve for x :

1. $\frac{2}{x+3} = \frac{1}{x-3}, x \neq \pm 3$

2. $\frac{2}{x+3} \leq \frac{1}{x-3}, x \neq \pm 3$

SOLUTION

Step 1: Solving the equation

To solve this equation we multiply both sides of the equation by $(x+3)(x-3)$ and simplify:

$$\frac{2}{x+3} \times (x+3)(x-3) = \frac{1}{x-3} \times (x+3)(x-3)$$

$$2(x-3) = x+3$$

$$2x - 6 = x + 3$$

$$x = 9$$

Step 2: Solving the inequality

It is very important to recognise that we cannot use the same method as above to solve the inequality. If we multiply or divide an inequality by a negative number, then the inequality sign changes direction. We must rather simplify the inequality to have a lowest common denominator and use a table of signs to determine the values that satisfy the inequality.

Step 3: Subtract $\frac{1}{x-3}$ from both sides of the inequality

$$\frac{2}{x+3} - \frac{1}{x-3} \leq 0$$

Step 4: Determine the lowest common denominator and simplify the fraction

$$\frac{2(x-3) - (x+3)}{(x+3)(x-3)} \leq 0$$

$$\frac{x-9}{(x+3)(x-3)} \leq 0$$

Keep the denominator because it affects the final answer.

Step 5: Determine the critical values of x

From the factorised inequality we see that the critical values are $x = -3$, $x = 3$ and $x = 9$.

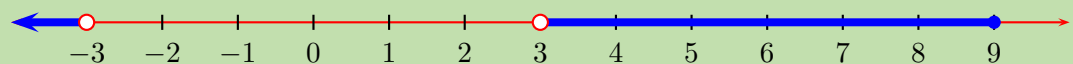
Step 6: Complete a table of signs

Critical values		$x = -3$		$x = 3$		$x = 9$	
$x + 3$	—	undef	+	+	+	+	+
$x - 3$	—	—	—	undef	+	+	+
$x - 9$	—	—	—	—	—	0	+
$f(x) = \frac{x-9}{(x+3)(x-3)}$	—	undef	+	undef	—	0	+

From the table we see that the function is less than or equal to zero for $x < -3$ or $3 < x \leq 9$. We do not include $x = -3$ or $x = 3$ in the solution because of the restrictions on the denominator.

Step 7: Write the final answer and represent on a number line

$$x < -3 \quad \text{or} \quad 3 < x \leq 9$$



Exercise 2 – 8: Solving quadratic inequalities

1. Solve the following inequalities and show each answer on a number line:

a) $x^2 - x < 12$

b) $3x^2 > -x + 4$

c) $y^2 < -y - 2$

d) $(3 - t)(1 + t) > 0$

e) $s^2 - 4s > -6$

f) $0 \geq 7x^2 - x + 8$

g) $x \geq -4x^2$

h) $2x^2 + x + 6 \leq 0$

i) $\frac{x}{x-3} < 2, x \neq 3$

j) $\frac{x^2 + 4}{x - 7} \geq 0, x \neq 7$

k) $\frac{x+2}{x} - 1 \geq 0, x \neq 0$

2. Draw a sketch of the following inequalities and solve for x :

a) $2x^2 - 18 > 0$

b) $5 - x^2 \leq 0$

c) $x^2 < 0$

d) $0 \geq 6x^2$

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1g. [22BH](#) 1h. [22BJ](#) 1i. [22BK](#) 1j. [22BM](#) 1k. [22BN](#) 2a. [22BP](#)
2b. [22BQ](#) 2c. [22BR](#) 2d. [22BS](#)



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2.8 Simultaneous equations

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Simultaneous linear equations can be solved using three different methods: substitution, elimination or using a graph to determine where the two lines intersect. For solving systems of simultaneous equations with linear and non-linear equations, we mostly use the substitution method. Graphical solution is useful for showing where the two equations intersect.

In general, to solve for the values of n unknown variables requires a system of n independent equations.

An example of a system of simultaneous equations with one linear equation and one quadratic equation is

$$y - 2x = -4$$

$$x^2 + y = 4$$

Solving by substitution

EMBFT

- Use the simplest of the two given equations to express one of the variables in terms of the other.
- Substitute into the second equation. By doing this we reduce the number of equations and the number of variables by one.

- We now have one equation with one unknown variable which can be solved.
- Use the solution to substitute back into the first equation to find the value of the other unknown variable.

Worked example 20: Simultaneous equations

QUESTION

Solve for x and y :

$$y - 2x = -4 \quad \dots (1)$$

$$x^2 + y = 4 \quad \dots (2)$$

SOLUTION

Step 1: Make y the subject of the first equation

$$y = 2x - 4$$

Step 2: Substitute into the second equation and simplify

$$x^2 + (2x - 4) = 4$$

$$x^2 + 2x - 8 = 0$$

Step 3: Factorise the equation

$$(x + 4)(x - 2) = 0$$

$$\therefore x = -4 \text{ or } x = 2$$

Step 4: Substitute the values of x back into the first equation to determine the corresponding y -values

If $x = -4$:

$$y = 2(-4) - 4$$

$$= -12$$

If $x = 2$:

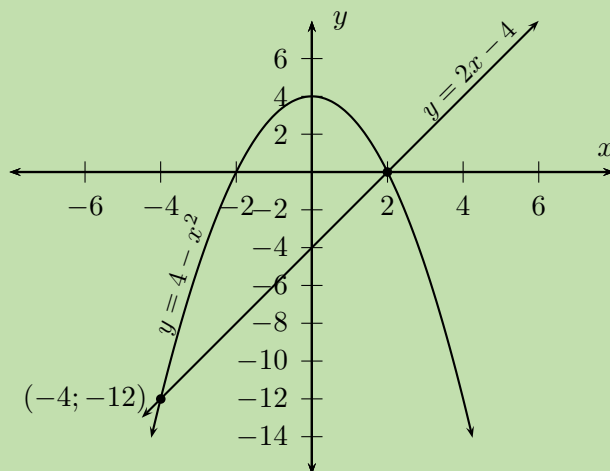
$$y = 2(2) - 4$$

$$= 0$$

Step 5: Check that the two points satisfy both original equations

Step 6: Write the final answer

The solution is $x = -4$ and $y = -12$ or $x = 2$ and $y = 0$. These are the coordinate pairs for the points of intersection as shown below.



Solving by elimination

EMBFV

- Make one of the variables the subject of both equations.
- Equate the two equations; by doing this we reduce the number of equations and the number of variables by one.
- We now have one equation with one unknown variable which can be solved.
- Use the solution to substitute back into either original equation, to find the corresponding value of the other unknown variable.

Worked example 21: Simultaneous equations

QUESTION

Solve for x and y :

$$\begin{aligned}y &= x^2 - 6x & \dots (1) \\y + \frac{1}{2}x - 3 &= 0 & \dots (2)\end{aligned}$$

SOLUTION

Step 1: Make y the subject of the second equation

$$\begin{aligned}y + \frac{1}{2}x - 3 &= 0 \\y &= -\frac{1}{2}x + 3\end{aligned}$$

Step 2: Equate the two equations and solve for x

$$\begin{aligned}x^2 - 6x &= -\frac{1}{2}x + 3 \\x^2 - 6x + \frac{1}{2}x - 3 &= 0 \\2x^2 - 12x + x - 6 &= 0 \\2x^2 - 11x - 6 &= 0 \\(2x + 1)(x - 6) &= 0 \\\text{Therefore } x &= -\frac{1}{2} \text{ or } x = 6\end{aligned}$$

Step 3: Substitute the values for x back into the second equation to calculate the corresponding y -values

$$\begin{aligned}\text{If } x &= -\frac{1}{2}: \\y &= -\frac{1}{2}\left(-\frac{1}{2}\right) + 3 \\\therefore y &= 3\frac{1}{4}\end{aligned}$$

This gives the point $\left(-\frac{1}{2}; 3\frac{1}{4}\right)$.

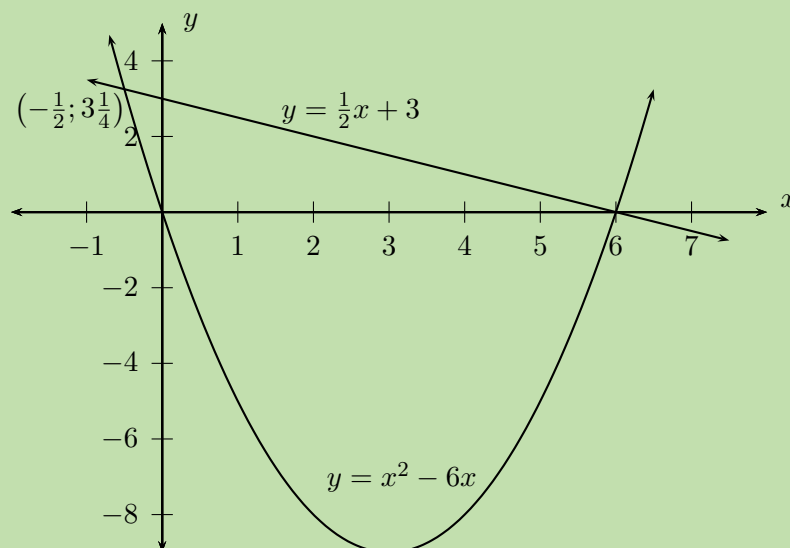
$$\begin{aligned}\text{If } x &= 6: \\y &= -\frac{1}{2}(6) + 3 \\&= -3 + 3 \\\therefore y &= 0\end{aligned}$$

This gives the point $(6; 0)$.

Step 4: Check that the two points satisfy both original equations

Step 5: Write the final answer

The solution is $x = -\frac{1}{2}$ and $y = 3\frac{1}{4}$ or $x = 6$ and $y = 0$. These are the coordinate pairs for the points of intersection as shown below.



Worked example 22: Simultaneous equations

QUESTION

Solve for x and y :

$$\begin{aligned}y &= \frac{5}{x-2} & \dots (1) \\y + 1 &= 2x & \dots (2)\end{aligned}$$

SOLUTION

Step 1: Make y the subject of the second equation

$$\begin{aligned}y + 1 &= 2x \\y &= 2x - 1\end{aligned}$$

Step 2: Equate the two equations and solve for x

$$\begin{aligned}2x - 1 &= \frac{5}{x-2} \\(2x - 1)(x - 2) &= 5 \\2x^2 - 5x + 2 &= 5 \\2x^2 - 5x - 3 &= 0 \\(2x + 1)(x - 3) &= 0 \\ \text{Therefore } x &= -\frac{1}{2} \text{ or } x = 3\end{aligned}$$

Step 3: Substitute the values for x back into the second equation to calculate the corresponding y -values

If $x = -\frac{1}{2}$:

$$\begin{aligned}y &= 2\left(-\frac{1}{2}\right) - 1 \\ \therefore y &= -2\end{aligned}$$

This gives the point $(-\frac{1}{2}; -2)$.

If $x = 3$:

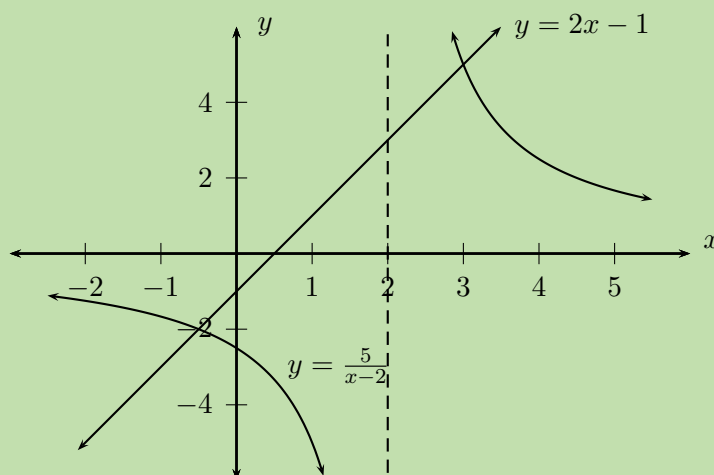
$$\begin{aligned}y &= 2(3) - 1 \\ &= 5\end{aligned}$$

This gives the point $(3; 5)$.

Step 4: Check that the two points satisfy both original equations

Step 5: Write the final answer

The solution is $x = -\frac{1}{2}$ and $y = -2$ or $x = 3$ and $y = 5$. These are the coordinate pairs for the points of intersection as shown below.



Solving graphically

EMBFW

- Make y the subject of each equation.
- Draw the graph of each equation on the same system of axes.
- The final solutions to the system of equations are the coordinates of the points where the two graphs intersect.

Worked example 23: Simultaneous equations

QUESTION

Solve graphically for x and y :

$$y + x^2 = 1 \quad \dots (1)$$

$$y - x + 5 = 0 \quad \dots (2)$$

SOLUTION

Step 1: Make y the subject of both equations

For the first equation we have

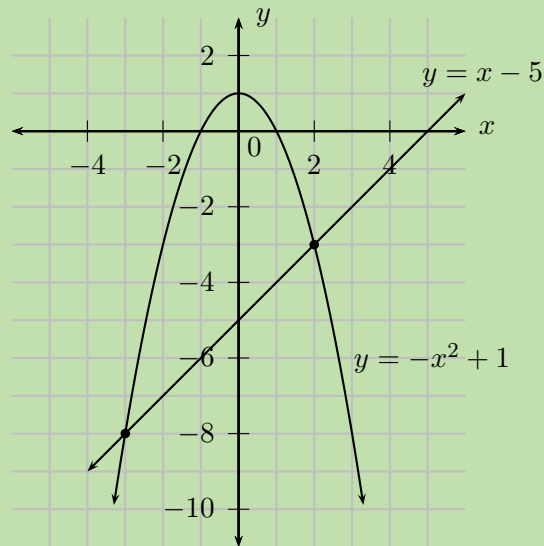
$$\begin{aligned} y + x^2 &= 1 \\ y &= -x^2 + 1 \end{aligned}$$

and for the second equation

$$y - x + 5 = 0$$

$$y = x - 5$$

Step 2: Draw the straight line graph and parabola on the same system of axes



Step 3: Determine where the two graphs intersect

From the diagram we see that the graphs intersect at $(-3; -8)$ and $(2; -3)$.

Step 4: Check that the two points satisfy both original equations

Step 5: Write the final answer

The solutions to the system of simultaneous equations are $(-3; -8)$ and $(2; -3)$.

Exercise 2 – 9: Solving simultaneous equations

1. Solve the following systems of equations algebraically. Leave your answer in surd form, where appropriate.

a) $y + x = 5$

$$y - x^2 + 3x - 5 = 0$$

b) $y = 6 - 5x + x^2$

$$y - x + 1 = 0$$

c) $y = \frac{2x + 2}{4}$

$$y - 2x^2 + 3x + 5 = 0$$

d) $a - 2b - 3 = 0; a - 3b^2 + 4 = 0$

e) $x^2 - y + 2 = 3x$

$$4x = 8 + y$$

f) $2y + x^2 + 25 = 7x$

$$3x = 6y + 96$$

2. Solve the following systems of equations graphically. Check your solutions by also solving algebraically.

a) $x^2 - 1 - y = 0$

$y + x - 5 = 0$

b) $x + y - 10 = 0$

$x^2 - 2 - y = 0$

c) $xy = 12$

$7 = x + y$

d) $6 - 4x - y = 0$

$12 - 2x^2 - y = 0$

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2.9 Word problems

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Solving word problems requires using mathematical language to describe real-life contexts. Problem-solving strategies are often used in the natural sciences and engineering disciplines (such as physics, biology, and electrical engineering) but also in the social sciences (such as economics, sociology and political science). To solve word problems we need to write a set of equations that describes the problem mathematically.

Examples of real-world problem solving applications are:

- modelling population growth;
- modelling effects of air pollution;
- modelling effects of global warming;
- computer games;
- in the sciences, to understand how the natural world works;
- simulators that are used to train people in certain jobs, such as pilots, doctors and soldiers;
- in medicine, to track the progress of a disease.

▶ See video: [22C6](#) at www.everythingmaths.co.za

1. Read the problem carefully.
2. What is the question and what do we need to solve for?
3. Assign variables to the unknown quantities, for example, x and y .
4. Translate the words into algebraic expressions by rewriting the given information in terms of the variables.
5. Set up a system of equations.
6. Solve for the variables using substitution.
7. Check the solution.
8. Write the final answer.

Investigation: Simple word problems

Write an equation that describes the following real-world situations mathematically:

1. Mohato and Lindiwe both have colds. Mohato sneezes twice for each sneeze of Lindiwe's. If Lindiwe sneezes x times, write an equation describing how many times they both sneezed.
2. The difference of two numbers is 10 and the sum of their squares is 50. Find the two numbers.
3. Liboko builds a rectangular storeroom. If the diagonal of the room is $\sqrt{1312}$ m and the perimeter is 80 m, determine the dimensions of the room.
4. It rains half as much in July as it does in December. If it rains y mm in July, write an expression relating the rainfall in July and December.
5. Zane can paint a room in 4 hours. Tlali can paint a room in 2 hours. How long will it take both of them to paint a room together?
6. 25 years ago, Arthur was 5 years more than a third of Bongani's age. Today, Bongani is 26 years less than twice Arthur's age. How old is Bongani?
7. The product of two integers is 95. Find the integers if their total is 24.

Worked example 24: Gym membership

QUESTION

The annual gym subscription for a single member is R 1000, while an annual family membership is R 1500. The gym is considering increasing all membership fees by the same amount. If this is done then a single membership would cost $\frac{5}{7}$ of a family membership. Determine the amount of the proposed increase.

SOLUTION

Step 1: Identify the unknown quantity and assign a variable

Let the amount of the proposed increase be x .

Step 2: Use the given information to complete a table

	now	after increase
single	1000	$1000 + x$
family	1500	$1500 + x$

Step 3: Set up an equation

$$1000 + x = \frac{5}{7}(1500 + x)$$

Step 4: Solve for x

$$\begin{aligned}7000 + 7x &= 7500 + 5x \\2x &= 500 \\x &= 250\end{aligned}$$

Step 5: Write the final answer

The proposed increase is R 250.

Worked example 25: Corner coffee house

QUESTION

Erica has decided to treat her friends to coffee at the Corner Coffee House. Erica paid R 54,00 for four cups of cappuccino and three cups of filter coffee. If a cup of cappuccino costs R 3,00 more than a cup of filter coffee, calculate how much a cup of each type of coffee costs?

SOLUTION

Step 1: Method 1: identify the unknown quantities and assign two variables

Let the cost of a cappuccino be x and the cost of a filter coffee be y .

Step 2: Use the given information to set up a system of equations

$$4x + 3y = 54 \quad \dots (1)$$

$$x = y + 3 \quad \dots (2)$$

Step 3: Solve the equations by substituting the second equation into the first equation

$$4(y + 3) + 3y = 54$$

$$4y + 12 + 3y = 54$$

$$7y = 42$$

$$y = 6$$

If $y = 6$, then using the second equation we have

$$x = y + 3$$

$$= 6 + 3$$

$$= 9$$

Step 4: Check that the solution satisfies both original equations

Step 5: Write the final answer

A cup of cappuccino costs R 9 and a cup of filter coffee costs R 6.

Step 6: Method 2: identify the unknown quantities and assign one variable

Let the cost of a cappuccino be x and the cost of a filter coffee be $x - 3$.

Step 7: Use the given information to set up an equation

$$4x + 3(x - 3) = 54$$

Step 8: Solve for x

$$4x + 3(x - 3) = 54$$

$$4x + 3x - 9 = 54$$

$$7x = 63$$

$$x = 9$$

Step 9: Write the final answer

A cup of cappuccino costs R 9 and a cup of filter coffee costs R 6.

Worked example 26: Taps filling a container

QUESTION

Two taps, one more powerful than the other, are used to fill a container. Working on its own, the less powerful tap takes 2 hours longer than the other tap to fill the container. If both taps are opened, it takes 1 hour, 52 minutes and 30 seconds to fill the container. Determine how long it takes the less powerful tap to fill the container on its own.

SOLUTION

Step 1: Identify the unknown quantities and assign variables

Let the time taken for the less powerful tap to fill the container be x and let the time taken for the more powerful tap be $x - 2$.

Step 2: Convert all units of time to be the same

First we must convert 1 hour, 52 minutes and 30 seconds to hours:

$$1 + \frac{52}{60} + \frac{30}{(60)^2} = 1,875 \text{ hours}$$

Step 3: Use the given information to set up a system of equations

Write an equation describing the two taps working together to fill the container:

$$\frac{1}{x} + \frac{1}{x - 2} = \frac{1}{1,875}$$

Step 4: Multiply the equation through by the lowest common denominator and simplify

$$\begin{aligned} 1,875(x - 2) + 1,875x &= x(x - 2) \\ 1,875x - 3,75 + 1,875x &= x^2 - 2x \\ 0 &= x^2 - 5,75x + 3,75 \end{aligned}$$

Multiply the equation through by 4 to make it easier to factorise (or use the quadratic formula)

$$\begin{aligned} 0 &= 4x^2 - 23x + 15 \\ 0 &= (4x - 3)(x - 5) \end{aligned}$$

Therefore $x = \frac{3}{4}$ or $x = 5$.

We have calculated that the less powerful tap takes $\frac{3}{4}$ hours or 5 hours to fill the container, but we know that when both taps are opened it takes 1,875 hours. We can therefore discard the first solution $x = \frac{3}{4}$ hours.

So the less powerful tap fills the container in 5 hours and the more powerful tap takes 3 hours.

Step 5: Check that the solution satisfies the original equation

Step 6: Write the final answer

The less powerful tap fills the container in 5 hours and the more powerful tap takes 3 hours.

Exercise 2 – 10:

- Mr. Tsilatsila builds a fence around his rectangular vegetable garden of 8 m^2 . If the length is twice the breadth, determine the dimensions of Mr. Tsilatsila's vegetable garden.
- Kevin has played a few games of ten-pin bowling. In the third game, Kevin scored 80 more than in the second game. In the first game Kevin scored 110 less than the third game. His total score for the first two games was 208. If he wants an average score of 146, what must he score on the fourth game?

- When an object is dropped or thrown downward, the distance, d , that it falls in time, t , is described by the following equation:

$$s = 5t^2 + v_0t$$

In this equation, v_0 is the initial velocity, in $\text{m}\cdot\text{s}^{-1}$. Distance is measured in meters and time is measured in seconds. Use the equation to find how long it takes a tennis ball to reach the ground if it is thrown downward from a hot-air balloon that is 500 m high. The tennis ball is thrown at an initial velocity of $5 \text{ m}\cdot\text{s}^{-1}$.

- The table below lists the times that Sheila takes to walk the given distances.

time (minutes)	5	10	15	20	25	30
distance (km)	1	2	3	4	5	6

Plot the points.

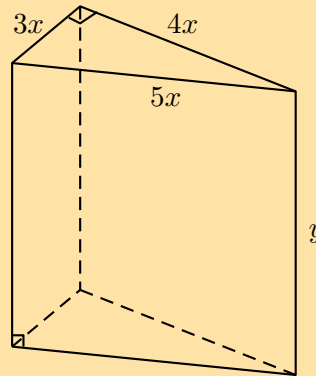
Find the equation that describes the relationship between time and distance. Then use the equation to answer the following questions:

- How long will it take Sheila to walk 21 km?
- How far will Sheila walk in 7 minutes?

If Sheila were to walk half as fast as she is currently walking, what would the graph of her distances and times look like?

- The power P (in watts) supplied to a circuit by a 12 volt battery is given by the formula $P = 12I - 0,5I^2$ where I is the current in amperes.
 - Since both power and current must be greater than 0, find the limits of the current that can be drawn by the circuit.

- b) Draw a graph of $P = 12I - 0,5I^2$ and use your answer to the first question to define the extent of the graph.
- c) What is the maximum current that can be drawn?
- d) From your graph, read off how much power is supplied to the circuit when the current is 10 A. Use the equation to confirm your answer.
- e) At what value of current will the power supplied be a maximum?
6. A wooden block is made as shown in the diagram. The ends are right-angled triangles having sides $3x$, $4x$ and $5x$. The length of the block is y . The total surface area of the block is 3600 cm^2 .



Show that

$$y = \frac{300 - x^2}{x}$$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

1. 22C7 2. 22C8 3. 22C9 4. 22CB 5. 22CC 6. 22CD



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2.10 Summary

EMBFZ

▶ See presentation: 22CF at www.everythingmaths.co.za

- Zero product law: if $a \times b = 0$, then $a = 0$ and/or $b = 0$.
- Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
- Discriminant: $\Delta = b^2 - 4ac$

Nature of roots	Discriminant
Roots are non-real	$\Delta < 0$
Roots are real and equal	$\Delta = 0$
Roots are real and unequal:	$\Delta > 0$
– Rational roots – Irrational roots	– $\Delta = \text{squared rational number}$ – $\Delta = \text{not squared rational number}$

Exercise 2 – 11: End of chapter exercises

- Solve: $x^2 - x - 1 = 0$. Give your answer correct to two decimal places.
- Solve: $16(x + 1) = x^2(x + 1)$
- Solve: $y^2 + 3 + \frac{12}{y^2 + 3} = 7$
- Solve for x : $2x^4 - 5x^2 - 12 = 0$
- Solve for x :
 - $x(x - 9) + 14 = 0$
 - $x^2 - x = 3$ (correct to one decimal place)
 - $x + 2 = \frac{6}{x}$ (correct to two decimal places)
 - $\frac{1}{x+1} + \frac{2x}{x-1} = 1$
- Solve for x in terms of p by completing the square: $x^2 - px - 4 = 0$
- The equation $ax^2 + bx + c = 0$ has roots $x = \frac{2}{3}$ and $x = -4$. Find one set of possible values for a , b and c .
- The two roots of the equation $4x^2 + px - 9 = 0$ differ by 5. Calculate the value of p .
- An equation of the form $x^2 + bx + c = 0$ is written on the board. Saskia and Sven copy it down incorrectly. Saskia has a mistake in the constant term and obtains the solutions -4 and 2 . Sven has a mistake in the coefficient of x and obtains the solutions 1 and -15 . Determine the correct equation that was on the board.
- For which values of b will the expression $\frac{b^2 - 5b + 6}{b + 2}$ be:
 - undefined?
 - equal to zero?
- Given $\frac{(x^2 - 6)(2x + 1)}{x + 2} = 0$ solve for x if:
 - x is a real number.
 - x is a rational number.
 - x is an irrational number.
 - x is an integer.

12. Given $\frac{(x-6)^{\frac{1}{2}}}{x^2+3}$, for which value(s) of x will the expression be:

- a) equal to zero?
- b) defined?

13. Solve for a if $\frac{\sqrt{8-2a}}{a-3} \geq 0$.

14. Abdoul stumbled across the following formula to solve the quadratic equation $ax^2 + bx + c = 0$ in a foreign textbook.

$$x = \frac{2c}{-b \pm \sqrt{b^2 - 4ac}}$$

- a) Use this formula to solve the equation: $2x^2 + x - 3 = 0$.
- b) Solve the equation again, using factorisation, to see if the formula works for this equation.
- c) Trying to derive this formula to prove that it always works, Abdoul got stuck along the way. His attempt is shown below:

$$ax^2 + bx + c = 0$$

$$a + \frac{b}{x} + \frac{c}{x^2} = 0 \quad \text{Divided by } x^2 \text{ where } x \neq 0$$

$$\frac{c}{x^2} + \frac{b}{x} + a = 0 \quad \text{Rearranged}$$

$$\frac{1}{x^2} + \frac{b}{cx} + \frac{a}{c} = 0 \quad \text{Divided by } c \text{ where } c \neq 0$$

$$\frac{1}{x^2} + \frac{b}{cx} = -\frac{a}{c} \quad \text{Subtracted } \frac{a}{c} \text{ from both sides}$$

$$\therefore \frac{1}{x^2} + \frac{b}{cx} + \dots \quad \text{Got stuck}$$

Complete his derivation.

15. Solve for x :

$$\text{a) } \frac{4}{x-3} \leq 1$$

$$\text{f) } 2x \leq \frac{15-x}{x}$$

$$\text{b) } \frac{4}{(x-3)^2} < 1$$

$$\text{g) } \frac{x^2+3}{3x-2} \leq 0$$

$$\text{c) } \frac{2x-2}{x-3} > 3$$

$$\text{h) } x-2 \geq \frac{3}{x}$$

$$\text{d) } \frac{-3}{(x-3)(x+1)} < 0$$

$$\text{i) } \frac{x^2+3x-4}{5+x^4} \leq 0$$

$$\text{e) } (2x-3)^2 < 4$$

$$\text{j) } \frac{x-2}{3-x} \geq 1$$

16. Solve the following systems of equations algebraically. Leave your answer in surd form, where appropriate.

a) $y - 2x = 0$

$y - x^2 - 2x + 3 = 0$

b) $a - 3b = 0$

$a - b^2 + 4 = 0$

c) $y - x^2 - 5x = 0$

$10 = y - 2x$

d) $p = 2p^2 + q - 3$

$p - 3q = 1$

e) $a - b^2 = 0$

$a - 3b + 1 = 0$

f) $a - 2b + 1 = 0$

$a - 2b^2 - 12b + 4 = 0$

g) $y + 4x - 19 = 0$

$8y + 5x^2 - 101 = 0$

h) $a + 4b - 18 = 0$

$2a + 5b^2 - 57 = 0$

17. Solve the following systems of equations graphically:

a) $2y + x - 2 = 0$

$8y + x^2 - 8 = 0$

b) $y + 3x - 6 = 0$

$y = x^2 + 4 - 4x$

18. A stone is thrown vertically upwards and its height (in metres) above the ground at time t (in seconds) is given by:

$$h(t) = 35 - 5t^2 + 30t$$

Find its initial height above the ground.

19. After doing some research, a transport company has determined that the rate at which petrol is consumed by one of its large carriers, travelling at an average speed of x km per hour, is given by:

$$P(x) = \frac{55}{2x} + \frac{x}{200} \quad \text{litres per kilometre}$$

Assume that the petrol costs R 4,00 per litre and the driver earns R 18,00 per hour of travel time. Now deduce that the total cost, C , in Rands, for a 2000 km trip is given by:

$$C(x) = \frac{256\,000}{x} + 40x$$

20. Solve the following quadratic equations by either factorisation, completing the square or by using the quadratic formula:

- Always try to factorise first, then use the formula if the trinomial cannot be factorised.
- Solve some of the equations by completing the square.

a) $-4y^2 - 41y - 45 = 0$

b) $16x^2 + 20x = 36$

c) $42p^2 + 104p + 64 = 0$

d) $21y + 3 = 54y^2$

e) $36y^2 + 44y + 8 = 0$

f) $12y^2 - 14 = 22y$

g) $16y^2 + 0y - 81 = 0$

h) $3y^2 + 10y - 48 = 0$

i) $63 - 5y^2 = 26y$

j) $2x^2 - 30 = 2$

k) $2y^2 = 98$

21. One root of the equation $9y^2 + 32 = ky$ is 8. Determine the value of k and the other root.
22. a) Solve for x in $x^2 - x = 6$.
b) Hence, solve for y in $(y^2 - y)^2 - (y^2 - y) - 6 = 0$.
23. Solve for x : $x = \sqrt{8 - x} + 2$
24. a) Solve for y in $-4y^2 + 8y - 3 = 0$.
b) Hence, solve for p in $4(p - 3)^2 - 8(p - 3) + 3 = 0$.
25. Solve for x : $2(x + 3)^{\frac{1}{2}} = 9$
26. a) Without solving the equation $x + \frac{1}{x} = 3$, determine the value of $x^2 + \frac{1}{x^2}$.
b) Now solve $x + \frac{1}{x} = 3$ and use the result to assess the answer obtained in the question above.
27. Solve for y : $5(y - 1)^2 - 5 = 19 - (y - 1)^2$
28. Solve for t : $2t(t - \frac{3}{2}) = \frac{3}{2t^2 - 3t} + 2$

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- | | | | | | |
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| 1. 22CG | 2. 22CH | 3. 22CJ | 4. 22CK | 5a. 22CM | 5b. 22CN |
| 5c. 22CP | 5d. 22CQ | 6. 22CR | 7. 22CS | 8. 22CT | 9. 22CV |
| 10. 22CW | 11. 22CX | 12. 22CY | 13. 22CZ | 14. 22D2 | 15a. 22D3 |
| 15b. 22D4 | 15c. 22D5 | 15d. 22D6 | 15e. 22D7 | 15f. 22D8 | 15g. 22D9 |
| 15h. 22DB | 15i. 22DC | 15j. 22DD | 16a. 22DF | 16b. 22DG | 16c. 22DH |
| 16d. 22DJ | 16e. 22DK | 16f. 22DM | 16g. 22DN | 16h. 22DP | 17a. 22DQ |
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| 20d. 22DY | 20e. 22DZ | 20f. 22F2 | 20g. 22F3 | 20h. 22F4 | 20i. 22F5 |
| 20j. 22F6 | 20k. 22F7 | 21. 22F8 | 22. 22F9 | 23. 22FB | 24. 22FC |
| 25. 22FD | 26. 22FF | 27. 22FG | 28. 22FH | | |



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Number patterns

3.1	<i>Revision</i>	86
3.2	<i>Quadratic sequences</i>	90
3.3	<i>Summary</i>	99

In earlier grades we learned about linear sequences, where the difference between consecutive terms is constant. In this chapter, we will learn about quadratic sequences, where the difference between consecutive terms is not constant, but follows its own pattern.

3.1 Revision

EMBG2

Terminology:	
Sequence/pattern	A sequence or pattern is an ordered set of numbers or variables.
Successive/consecutive	Successive or consecutive terms are terms that directly follow one after another in a sequence.
Common difference	The common or constant difference (d) is the difference between any two consecutive terms in a linear sequence.
General term	A mathematical expression that describes the sequence and that generates any term in the pattern by substituting different values for n .
Conjecture	A statement, consistent with known data, that has not been proved true nor shown to be false.

Important: a series is not the same as a sequence or pattern. Different types of series are studied in Grade 12. In Grade 11 we study sequences only.

▶ See video: 22FJ at www.everythingmaths.co.za

Describing patterns

EMBG3

To describe terms in a pattern we use the following notation:

- T_1 is the first term of a sequence.
- T_4 is the fourth term of a sequence.
- T_n is the general term and is often expressed as the n^{th} term of a sequence.

A sequence does not have to follow a pattern but when it does, we can write an equation for the general term. The general term can be used to calculate any term in the sequence. For example, consider the following linear sequence: 1; 4; 7; 10; 13; ... The n^{th} term is given by the equation $T_n = 3n - 2$.

You can check this by substituting values for n :

$$T_1 = 3(1) - 2 = 1$$

$$T_2 = 3(2) - 2 = 4$$

$$T_3 = 3(3) - 2 = 7$$

$$T_4 = 3(4) - 2 = 10$$

$$T_5 = 3(5) - 2 = 13$$

If we find the relationship between the position of a term and its value, we can describe the pattern and find any term in the sequence.

► See video: 22FK at www.everythingmaths.co.za

Linear sequences

EMBG4

DEFINITION: Linear sequence

A sequence of numbers in which there is a common difference (d) between any term and the term before it is called a linear sequence.

Important: $d = T_2 - T_1$, not $T_1 - T_2$.

Worked example 1: Linear sequence

QUESTION

Determine the common difference (d) and the general term for the following sequence:

10; 7; 4; 1; ...

SOLUTION

Step 1: Determine the common difference

To calculate the common difference, we find the difference between any term and the previous term:

$$d = T_n - T_{n-1}$$

$$\text{Therefore } d = T_2 - T_1$$

$$= 7 - 10$$

$$= -3$$

$$\text{or } d = T_3 - T_2$$

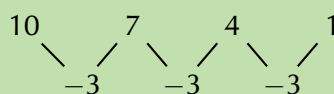
$$= 4 - 7$$

$$= -3$$

$$\text{or } d = T_4 - T_3$$

$$= 1 - 4$$

$$= -3$$



Step 2: Determine the general term

To find the general term T_n , we must identify the relationship between:

- the **value** of a number in the pattern and
- the **position** of a number in the pattern

position	1	2	3	4
value	10	7	4	1

We start with the value of the first term in the sequence. We need to write an expression that includes the value of the common difference ($d = -3$) and the position of the term ($n = 1$).

$$\begin{aligned}
 T_1 &= 10 \\
 &= 10 + (0)(-3) \\
 &= 10 + (1 - 1)(-3)
 \end{aligned}$$

Now we write a similar expression for the second term.

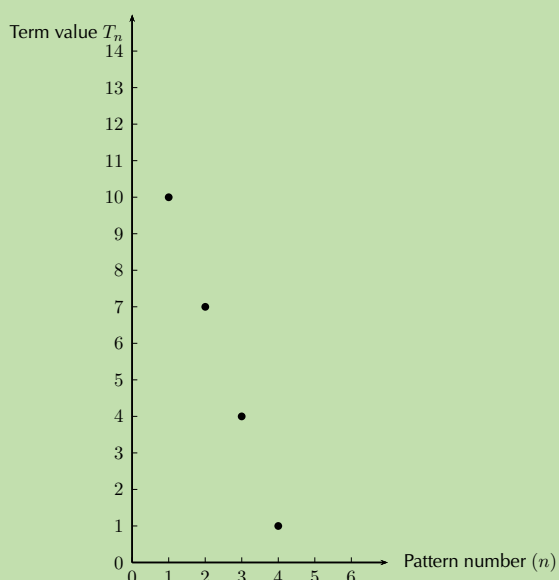
$$\begin{aligned}
 T_2 &= 7 \\
 &= 10 + (1)(-3) \\
 &= 10 + (2 - 1)(-3)
 \end{aligned}$$

We notice a pattern forming that links the position of a number in the sequence to its value.

$$\begin{aligned}
 T_n &= 10 + (n - 1)(-3) \\
 &= 10 - 3n + 3 \\
 &= -3n + 13
 \end{aligned}$$

Step 3: Drawing a graph of the pattern

We can also represent this pattern graphically, as shown below.



Notice that the position numbers (n) can be positive integers only.

This pattern can also be expressed in words: “each term in the sequence can be calculated by multiplying negative three and the position number, and then adding thirteen.”

Exercise 3 – 1: Linear sequences

- Write down the next three terms in each of the following sequences:
45; 29; 13; -3; ...
- The general term is given for each sequence below. Calculate the missing terms.
 - 4; -9; -14; ...; -24
 $T_n = 1 - 5n$
 - 6; ...; 24; ...; 42
 $T_n = 9n - 3$
- Find the general formula for the following sequences and then find T_{10} , T_{15} and T_{30} :
 - 13; 16; 19; 22; ...
 - 18; 24; 30; 36; ...
 - 10; -15; -20; -25; ...
- The seating in a classroom is arranged so that the first row has 20 desks, the second row has 22 desks, the third row has 24 desks and so on. Calculate how many desks are in the ninth row.
- Complete the following:

$$13 + 31 = \dots$$

$$24 + 42 = \dots$$

$$38 + 83 = \dots$$

- Look at the numbers on the left-hand side, what do you notice about the unit digit and the tens-digit?
- Investigate the pattern by trying other examples of 2-digit numbers.
- Make a conjecture about the pattern that you notice.
- Prove this conjecture.

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1. [22FN](#) 2a. [22FP](#) 2b. [22FQ](#) 3a. [22FR](#) 3b. [22FS](#) 3c. [22FT](#)
4. [22FV](#) 5. [22FW](#)

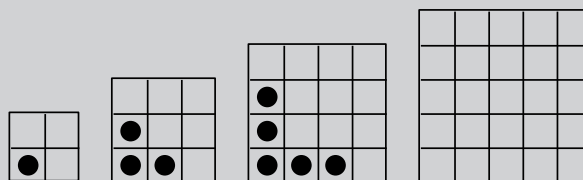


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Investigation: Quadratic sequences



1. Study the dotted-tile pattern shown and answer the following questions.

- Complete the fourth pattern in the diagram.
- Complete the table below:

pattern number	1	2	3	4	5	20	n
dotted tiles	1	3	5				
difference (d)	—	2					

- What do you notice about the change in number of dotted tiles?
 - Describe the pattern in words: “The number of dotted tiles...”.
 - Write the general term: $T_n = \dots$
 - Give the mathematical name for this kind of pattern.
 - A pattern has 819 dotted tiles. Determine the value of n .
2. Now study the number of blank tiles (tiles without dots) and answer the following questions:

- Complete the table below:

pattern number	1	2	3	4	5	10
blank tiles	3	6	11			
first difference	—	3				
second difference	—	—				

- What do you notice about the change in the number of blank tiles?
- Describe the pattern in words: “The number of blank tiles...”.
- Write the general term: $T_n = \dots$
- Give the mathematical name for this kind of pattern.
- A pattern has 227 blank tiles. Determine the value of n .
- A pattern has 79 dotted tiles. Determine the number of blank tiles.

DEFINITION: Quadratic sequence

A quadratic sequence is a sequence of numbers in which the second difference between any two consecutive terms is constant.

Consider the following example: 1; 2; 4; 7; 11; ...

The first difference is calculated by finding the difference between consecutive terms:

$$\begin{array}{ccccccc} 1 & & 2 & & 4 & & 7 & & 11 \\ & \searrow & \nearrow & \searrow & \nearrow & \searrow & \nearrow & \searrow & \nearrow \\ & +1 & & +2 & & +3 & & +4 & \end{array}$$

The second difference is obtained by taking the difference between consecutive first differences:

$$\begin{array}{ccccccc} 1 & & 2 & & 3 & & 4 \\ & \searrow & \nearrow & \searrow & \nearrow & \searrow & \nearrow \\ & +1 & & +1 & & +1 & \end{array}$$

We notice that the second differences are all equal to 1. Any sequence that has a common second difference is a **quadratic sequence**.

It is important to note that the first differences of a quadratic sequence form a sequence. This sequence has a constant difference between consecutive terms. In other words, a linear sequence results from taking the first differences of a quadratic sequence.

General case

If the sequence is quadratic, the n^{th} term is of the form $T_n = an^2 + bn + c$.

$$\begin{array}{ccccccc} & n=1 & & n=2 & & n=3 & & n=4 \\ T_n & a+b+c & & 4a+2b+c & & 9a+3b+c & & 16a+4b+c \\ 1^{\text{st}} \text{ difference} & & 3a+b & & 5a+b & & 7a+b & \\ 2^{\text{nd}} \text{ difference} & & & 2a & & 2a & & \end{array}$$

In each case, the common second difference is a $2a$.

Exercise 3 – 2: Quadratic sequences

1. Determine the second difference between the terms for the following sequences:

- | | |
|-----------------------------|--------------------------------------|
| a) 5; 20; 45; 80; ... | g) $-1; 2; 9; 20; \dots$ |
| b) 6; 11; 18; 27; ... | h) $1; -3; -9; -17; \dots$ |
| c) 1; 4; 9; 16; ... | i) $3a+1; 12a+1; 27a+1; 48a+1 \dots$ |
| d) 3; 0; $-5; -12; \dots$ | j) $2; 10; 24; 44; \dots$ |
| e) 1; 3; 7; 13; ... | k) $t-2; 4t-1; 9t; 16t+1; \dots$ |
| f) $0; -6; -16; -30; \dots$ | |

2. Complete the sequence by filling in the missing term:

- | | |
|------------------------|--------------------------|
| a) 11; 21; 35; ...; 75 | d) 3; ...; -13; -27; -45 |
| b) 20; ...; 42; 56; 72 | e) 24; 35; 48; ...; 80 |
| c) ...; 37; 65; 101 | f) ...; 11; 26; 47 |

3. Use the general term to generate the first four terms in each sequence:

- a) $T_n = n^2 + 3n - 1$
b) $T_n = -n^2 - 5$
c) $T_n = 3n^2 - 2n$
d) $T_n = -2n^2 + n + 1$

Think you got it? Get this answer and more practice on our Intelligent Practice Service

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|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| 1a. 22FX | 1b. 22FY | 1c. 22FZ | 1d. 22G2 | 1e. 22G3 | 1f. 22G4 |
| 1g. 22G5 | 1h. 22G6 | 1i. 22G7 | 1j. 22G8 | 1k. 22G9 | 2a. 22GB |
| 2b. 22GC | 2c. 22GD | 2d. 22GF | 2e. 22GG | 2f. 22GH | 3a. 22GJ |
| 3b. 22GK | 3c. 22GM | 3d. 22GN | | | |



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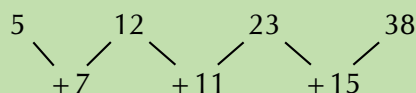
Worked example 2: Quadratic sequences

QUESTION

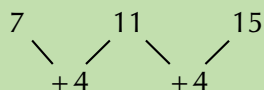
Write down the next two terms and determine an equation for the n^{th} term of the sequence 5; 12; 23; 38; ...

SOLUTION

Step 1: Find the first differences between the terms

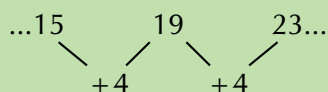


Step 2: Find the second differences between the terms



So there is a common second difference of 4. We can therefore conclude that this is a quadratic sequence of the form $T_n = an^2 + bn + c$.

Continuing the sequence, the next first differences will be:



Step 3: Finding the next two terms in the sequence

The next two terms will be:

$$\begin{array}{ccccc} \dots & 38 & & 57 & & 80 & \dots \\ & \swarrow & & \swarrow & & \swarrow & \\ & +19 & & +23 & & & \end{array}$$

So the sequence will be: 5; 12; 23; 38; 57; 80; ...

Step 4: Determine the general term for the sequence

To find the values of a , b and c for $T_n = an^2 + bn + c$ we look at the first 3 terms in the sequence:

$$n = 1 : T_1 = a + b + c$$

$$n = 2 : T_2 = 4a + 2b + c$$

$$n = 3 : T_3 = 9a + 3b + c$$

We solve a set of simultaneous equations to determine the values of a , b and c

We know that $T_1 = 5$, $T_2 = 12$ and $T_3 = 23$

$$a + b + c = 5$$

$$4a + 2b + c = 12$$

$$9a + 3b + c = 23$$

$$T_2 - T_1 = 4a + 2b + c - (a + b + c)$$

$$12 - 5 = 4a + 2b + c - a - b - c$$

$$7 = 3a + b \quad \dots (1)$$

$$T_3 - T_2 = 9a + 3b + c - (4a + 2b + c)$$

$$23 - 12 = 9a + 3b + c - 4a - 2b - c$$

$$11 = 5a + b \quad \dots (2)$$

$$(2) - (1) = 5a + b - (3a + b)$$

$$11 - 7 = 5a + b - 3a - b$$

$$4 = 2a$$

$$\therefore a = 2$$

$$\text{Using equation (1) : } 3(2) + b = 7$$

$$\therefore b = 1$$

$$\text{And using } a + b + c = 5$$

$$2 + 1 + c = 5$$

$$\therefore c = 1$$

Step 5: Write the general term for the sequence

$$T_n = 2n^2 + n + 2$$

Worked example 3: Plotting a graph of terms in a sequence

QUESTION

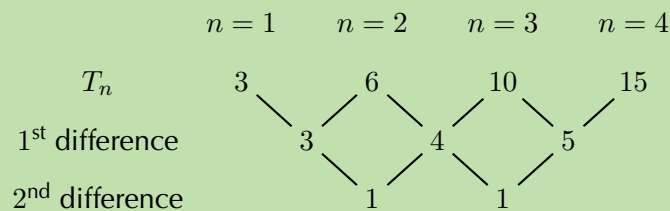
Consider the following sequence:

$$3; 6; 10; 15; 21; \dots$$

1. Determine the general term (T_n) for the sequence.
2. Is this a linear or a quadratic sequence?
3. Plot a graph of T_n vs n .

SOLUTION

Step 1: Determine the first and second differences



We see that the first differences are not constant and form the sequence $3; 4; 5; \dots$ and that there is a common second difference of 1. Therefore the sequence is quadratic and has a general term of the form $T_n = an^2 + bn + c$.

Step 2: Determine the general term T_n

To find the values of a , b and c for $T_n = an^2 + bn + c$ we look at the first 3 terms in the sequence:

$$n = 1 : T_1 = a + b + c$$

$$n = 2 : T_2 = 4a + 2b + c$$

$$n = 3 : T_3 = 9a + 3b + c$$

We solve this set of simultaneous equations to determine the values of a , b and c . We know that $T_1 = 3$, $T_2 = 6$ and $T_3 = 10$.

$$a + b + c = 3$$

$$4a + 2b + c = 6$$

$$9a + 3b + c = 10$$

$$T_2 - T_1 = 4a + 2b + c - (a + b + c)$$

$$6 - 3 = 4a + 2b + c - a - b - c$$

$$3 = 3a + b$$

... (1)

$$T_3 - T_2 = 9a + 3b + c - (4a + 2b + c)$$

$$10 - 6 = 9a + 3b + c - 4a - 2b - c$$

$$4 = 5a + b$$

... (2)

$$(2) - (1) = 5a + b - (3a + b)$$

$$4 - 3 = 5a + b - 3a - b$$

$$1 = 2a$$

$$\therefore a = \frac{1}{2}$$

$$\text{Using equation (1): } 3\left(\frac{1}{2}\right) + b = 3$$

$$\therefore b = \frac{3}{2}$$

$$\text{And using } a + b + c = 3$$

$$\frac{1}{2} + \frac{3}{2} + c = 3$$

$$\therefore c = 1$$

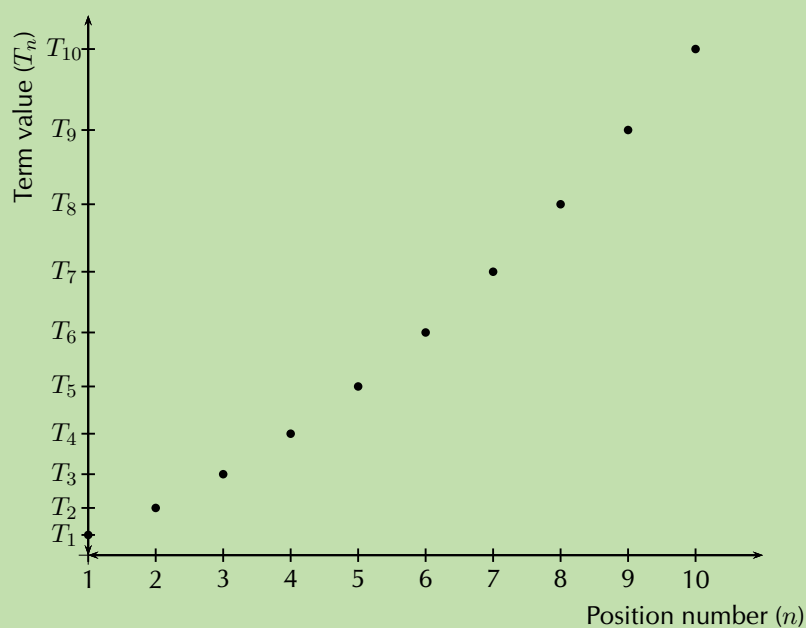
Therefore the general term for the sequence is $T_n = \frac{1}{2}n^2 + \frac{3}{2}n + 1$.

Step 3: Plot a graph of T_n vs n

Use the general term for the sequence, $T_n = \frac{1}{2}n^2 + \frac{3}{2}n + 1$, to complete the table.

n	1	2	3	4	5	6	7	8	9	10
T_n	3	6	10	15	21	28	36	45	55	66

Use the table to plot the graph:



In this case it would not be accurate to join these points, since n indicates the position of a term in a sequence and can therefore only be a positive integer. We can, however, see that the plot of the points lies in the shape of a parabola.

Worked example 4: Olympic Games soccer event

QUESTION

In the first stage of the soccer event at the Olympic Games, there are teams from four different countries in each group. Each country in a group must play every other country in the group once.

1. How many matches will be played in each group in the first stage of the event?
2. How many matches would be played if there are 5 teams in each group?
3. How many matches would be played if there are 6 teams in each group?
4. Determine the general formula of the sequence.

SOLUTION

Step 1: Determine the number of matches played if there are 4 teams in a group

Let the teams from four different countries be A , B , C and D .

teams in a group	matches played
A	AB, AC, AD
B	BC, BD
C	CD
D	
4	$3 + 2 + 1 = 6$

AB means that team A plays team B and BA would be the same match as AB . So if there are four different teams in a group, each group plays 6 matches.

Step 2: Determine the number of matches played if there are 5 teams in a group

Let the teams from five different countries be A , B , C , D and E .

teams in a group	matches played
A	AB, AC, AD, AE
B	BC, BD, BE
C	CD, CE
D	DE
E	
5	$4 + 3 + 2 + 1 = 10$

So if there are five different teams in a group, each group plays 10 matches.

Step 3: Determine the number of matches played if there are 6 teams in a group

Let the teams from six different countries be A , B , C , D , E and F .

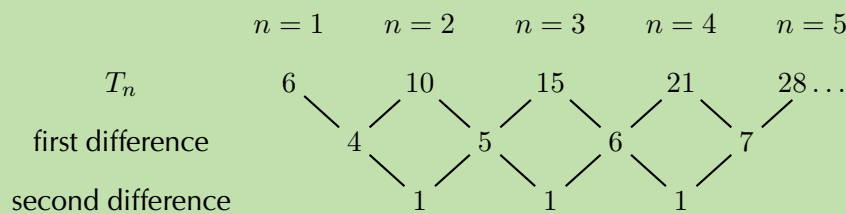
teams in a group	matches to be played
<i>A</i>	<i>AB, AC, AD, AE, AF</i>
<i>B</i>	<i>BC, BD, BE, BF</i>
<i>C</i>	<i>CD, CE, CF</i>
<i>D</i>	<i>DE, DF</i>
<i>E</i>	<i>EF</i>
<i>F</i>	
5	$5 + 4 + 3 + 2 + 1 = 15$

So if there are six different teams in a group, each group plays 15 matches.

We continue to increase the number of teams in a group and find that a group of 7 teams plays 21 matches and a group of 8 teams plays 28 matches.

Step 4: Consider the sequence

We examine the sequence to determine if it is linear or quadratic:



We see that the first differences are not constant and that there is a common second difference of 1. Therefore the sequence is quadratic and has a general term of the form $T_n = an^2 + bn + c$.

Step 5: Determine the general term T_n

To find the values of a , b and c for $T_n = an^2 + bn + c$ we look at the first 3 terms in the sequence:

$$\begin{aligned} n = 1 : T_1 &= a + b + c \\ n = 2 : T_2 &= 4a + 2b + c \\ n = 3 : T_3 &= 9a + 3b + c \end{aligned}$$

We solve a set of simultaneous equations to determine the values of a , b and c . We know that $T_1 = 6$, $T_2 = 10$ and $T_3 = 15$

$$a + b + c = 6$$

$$4a + 2b + c = 10$$

$$9a + 3b + c = 15$$

$$\begin{aligned} T_2 - T_1 &= 4a + 2b + c - (a + b + c) \\ 10 - 6 &= 4a + 2b + c - a - b - c \\ 4 &= 3a + b \end{aligned} \quad \dots (1)$$

$$\begin{aligned} T_3 - T_2 &= 9a + 3b + c - (4a + 2b + c) \\ 15 - 10 &= 9a + 3b + c - 4a - 2b - c \\ 5 &= 5a + b \end{aligned} \quad \dots (2)$$

$$(2) - (1) = 5a + b - (3a + b)$$

$$5 - 4 = 5a + b - 3a - b$$

$$1 = 2a$$

$$\therefore a = \frac{1}{2}$$

$$\text{Using equation (1): } 3\left(\frac{1}{2}\right) + b = 4$$

$$\therefore b = \frac{5}{2}$$

$$\text{And using } a + b + c = 6$$

$$\frac{1}{2} + \frac{5}{2} + c = 6$$

$$\therefore c = 3$$

Therefore the general term for the sequence is $T_n = \frac{1}{2}n^2 + \frac{5}{2}n + 3$.

Exercise 3 – 3: Quadratic sequences

1. Calculate the common second difference for each of the following quadratic sequences:

a) 3; 6; 10; 15; 21; ...

d) 2; 10; 26; 50; 82; ...

b) 4; 9; 16; 25; 36; ...

c) 7; 17; 31; 49; 71; ...

e) 31; 30; 27; 22; 15; ...

2. Find the first five terms of the quadratic sequence defined by: $T_n = 5n^2 + 3n + 4$.

3. Given $T_n = 4n^2 + 5n + 10$, find T_9 .

4. Given $T_n = 2n^2$, for which value of n does $T_n = 32$?

5. a) Write down the next two terms of the quadratic sequence: 16; 27; 42; 61; ...

b) Find the general formula for the quadratic sequence above.

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1a. 22GP

1b. 22GQ

1c. 22GR

1d. 22GS

1e. 22GT

2. 22GV

3. 22GW

4. 22GX

5. 22GY



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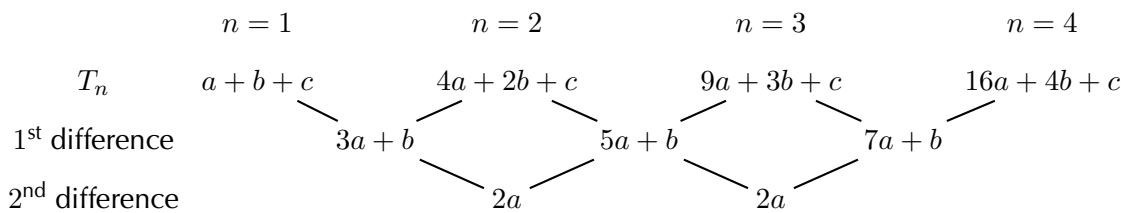
- T_n is the general term of a sequence.
- **Successive** or **consecutive** terms are terms that follow one after another in a sequence.
- A **linear** sequence has a common difference (d) between any two successive terms.

$$d = T_n - T_{n-1}$$

- A **quadratic** sequence has a common second difference between any two successive terms.
- The general term for a quadratic sequence is

$$T_n = an^2 + bn + c$$

- A general quadratic sequence:



Exercise 3 – 4: End of chapter exercises

- Find the first five terms of the quadratic sequence defined by:

$$T_n = n^2 + 2n + 1$$

- Determine whether each of the following sequences is:

- a linear sequence,
- a quadratic sequence,
- or neither.

a) 6; 9; 14; 21; 30; ...

b) 1; 7; 17; 31; 49; ...

c) 8; 17; 32; 53; 80; ...

d) 9; 26; 51; 84; 125; ...

e) 2; 20; 50; 92; 146; ...

f) 5; 19; 41; 71; 109; ...

g) 2; 6; 10; 14; 18; ...

h) 3; 9; 15; 21; 27; ...

i) 1; 2,5; 5; 8,5; 13; ...

j) 10; 24; 44; 70; 102; ...

k) $2\frac{1}{2}$; 6; $10\frac{1}{2}$; 16; $22\frac{1}{2}$; ...

l) $3p^2$; $6p^2$; $9p^2$; $12p^2$; $15p^2$; ...

m) $2k$; $8k$; $18k$; $32k$; $50k$; ...

3. Given the pattern: $16; x; 46; \dots$, determine the value of x if the pattern is linear.
4. Given $T_n = 2n^2$, for which value of n does $T_n = 242$?
5. Given $T_n = 3n^2$, find T_{11} .
6. Given $T_n = n^2 + 4$, for which value of n does $T_n = 85$?
7. Given $T_n = 4n^2 + 3n - 1$, find T_5 .
8. Given $T_n = \frac{3}{2}n^2$, for which value of n does $T_n = 96$?
9. For each of the following patterns, determine:
 - the next term in the pattern,
 - and the general term,
 - the tenth term in the pattern.
 - a) $3; 7; 11; 15; \dots$
 - b) $17; 12; 7; 2; \dots$
 - c) $\frac{1}{2}; 1; 1\frac{1}{2}; 2; \dots$
 - d) $a; a + b; a + 2b; a + 3b; \dots$
 - e) $1; -1; -3; -5; \dots$
10. For each of the following sequences, find the equation for the general term and then use the equation to find T_{100} .

- a) $4; 7; 12; 19; 28; \dots$
- b) $2; 8; 14; 20; 26; \dots$
- c) $7; 13; 23; 37; 55; \dots$
- d) $5; 14; 29; 50; 77; \dots$

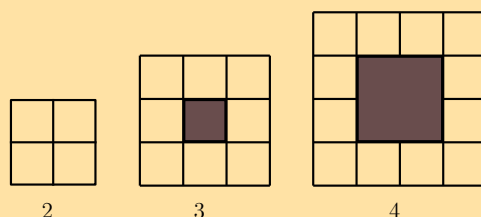
Given: $T_n = 3n - 1$

11. a) Write down the first five terms of the sequence.
- b) What do you notice about the difference between any two consecutive terms?
- c) Will this always be the case for a linear sequence?

Given the following sequence: $-15; -11; -7; \dots; 173$

12. a) Determine the equation for the general term.
- b) Calculate how many terms there are in the sequence.
13. Given $3; 7; 13; 21; 31; \dots$
 - a) Thabang determines that the general term is $T_n = 4n - 1$. Is he correct? Explain.
 - b) Cristina determines that the general term is $T_n = n^2 + n + 1$. Is she correct? Explain.

14. Given the following pattern of blocks:

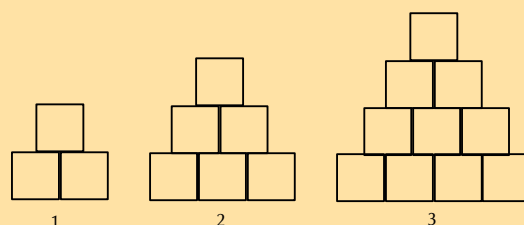


- a) Draw pattern 5.
b) Complete the table below:

pattern number (n)	2	3	4	5	10	250	n
number of white blocks (w)	4	8					

- c) Is this a linear or a quadratic sequence?

15. Cubes of volume 1 cm^3 are stacked on top of each other to form a tower:



- a) Complete the table for the height of the tower:

tower number (n)	1	2	3	4	10	n
height of tower (h)	2					

- b) What type of sequence is this?
c) Now consider the number of cubes in each tower and complete the table below:

tower number (n)	1	2	3	4
number of cubes (c)	3			

- d) What type of sequence is this?
e) Determine the general term for this sequence.
f) How many cubes are needed for tower number 21?
g) How high will a tower of 496 cubes be?

16. A quadratic sequence has a second term equal to 1, a third term equal to -6 and a fourth term equal to -14 .

- a) Determine the second difference for this sequence.
b) Hence, or otherwise, calculate the first term of the pattern.

17. There are 15 schools competing in the U16 girls hockey championship and every team must play two matches — one home match and one away match.

a) Use the given information to complete the table:

no. of schools	no. of matches
1	0
2	
3	
4	
5	

b) Calculate the second difference.

c) Determine a general term for the sequence.

d) How many matches will be played if there are 15 schools competing in the championship?

e) If 600 matches must be played, how many schools are competing in the championship?

18. The first term of a quadratic sequence is 4, the third term is 34 and the common second difference is 10. Determine the first six terms in the sequence.

19. Challenge question:

Given that the general term for a quadratic sequences is $T_n = an^2 + bn + c$, let d be the first difference and D be the second common difference.

a) Show that $a = \frac{D}{2}$.

b) Show that $b = d - \frac{3}{2}D$.

c) Show that $c = T_1 - d + D$.

d) Hence, show that $T_n = \frac{D}{2}n^2 + \left(d - \frac{3}{2}D\right)n + (T_1 - d + D)$.

Think you got it? Get this answer and more practice on our Intelligent Practice Service

- | | | | | | |
|--------------------------|---------------------------|---------------------------|---------------------------|---------------------------|--------------------------|
| 1. 22H2 | 2a. 22H3 | 2b. 22H4 | 2c. 22H5 | 2d. 22H6 | 2e. 22H7 |
| 2f. 22H8 | 2g. 22H9 | 2h. 22HB | 2i. 22HC | 2j. 22HD | 2k. 22HF |
| 2l. 22HG | 2m. 22HH | 3. 22HJ | 4. 22HK | 5. 22HM | 6. 22HN |
| 7. 22HP | 8. 22HQ | 9a. 22HR | 9b. 22HS | 9c. 22HT | 9d. 22HV |
| 9e. 22HW | 10a. 22HX | 10b. 22HY | 10c. 22HZ | 10d. 22J2 | 11. 22J3 |
| 12. 22J4 | 13. 22J5 | 14. 22J6 | 15. 22J7 | 16. 22J8 | 17. 22J9 |
| 18. 22JB | 19. 22JC | | | | |



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